

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4950	0.7739	0.8139	0.7979	0.8400	0.8800

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08556	0.13250	0.12569	0.16500	0.89944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08778	0.13556	0.12823	0.16778	0.76778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.015556	0.124022	0.146111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.006111	0.010556	0.082218	0.046389	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
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0.0005556 0.0233333 0.1727778 0.2483808 0.3650000 1.0000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5467	0.7072	0.7450	0.7354	0.7722	0.8083

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03944	0.08389	0.08811	0.12389	0.89556
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04111	0.08667	0.09049	0.12778	0.70278
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.030556	0.368214	0.911667	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0027778	0.2180843	0.2336111	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.07833	0.92333	0.64821	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5011	0.5722	0.6194	0.6240	0.6711	0.8156

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.05500	0.06336	0.09000	0.69278
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.05778	0.06564	0.09333	0.58222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2108392	0.4444444	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1850	0.4778	0.5272	0.5457	0.5928	0.9844

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.114591	0.007222	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02167	0.54556	0.47279	0.82278	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9622	0.9928	0.9950	0.9944	0.9967	1.0000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01611	0.03944	0.05051	0.07111	0.81333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.04111	0.05266	0.07444	0.61556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000e+00 0.000e+00 0.000e+00 1.577e-05 0.000e+00 1.389e-01
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 2.587e-06 0.000e+00 4.444e-03
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 7.440e-06 0.000e+00 3.889e-03
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 1.603e-05 0.000e+00 2.722e-01

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--- Summary for K = 5 ---

Z.ar:

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  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.9722 0.9928 0.9944 0.9941 0.9961 1.0000

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beta.ar:

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Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01111 0.02833 0.04040 0.05500 0.75389
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01222 0.03000 0.04211 0.05778 0.68444
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 7.930e-07 0.000e+00 8.333e-03
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 4.210e-06 0.000e+00 1.139e-01
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0 0 0 0 0 0
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 4.310e-06 0.000e+00 1.278e-01

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--- Summary for K = 6 ---

Z.ar:

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  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.9689 0.9922 0.9939 0.9936 0.9956 0.9994

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.021667	0.032911	0.043889	0.831667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.023333	0.034390	0.046667	0.466111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000077	0.0000000	0.3988889

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.550e-08	0.000e+00	5.556e-04

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.210e-07	0.000e+00	3.333e-03

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5306	0.7811	0.8239	0.8123	0.8517	0.9161

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.016111	0.025303	0.033889	0.593333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.017778	0.026678	0.036667	0.397778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0e+00	0.0e+00	0.0e+00	8.4e-05	0.0e+00	1.0e+00

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1589	0.2956	0.5744	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002912	0.0000000	1.0000000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9683	0.9911	0.9933	0.9931	0.9950	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013889	0.024402	0.030556	0.786667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.014444	0.025540	0.033333	0.424444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	6.900e-07	0.000e+00	3.889e-02

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 1.99e-05 0.00e+00 4.45e-01

```

--- Summary for K = 9 ---

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Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.9833 0.9917 0.9933 0.9932 0.9950 0.9989

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```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.010556 0.017914 0.022778 0.797778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

```

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gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.011111 0.018991 0.024444 0.330000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 1.76e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    0      0      0      0      0      0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 9.389e-05 0.000e+00 1.472e-01
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    0      0      0      0      0      0

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6067	0.8283	0.8433	0.8311	0.8522	0.8728

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08833	0.13389	0.12909	0.16667	0.86611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09056	0.13667	0.13173	0.17000	0.72333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.009444	0.104215	0.047778	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0083333	0.0776651	0.0262500	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
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0.000000 0.008333 0.036667 0.182825 0.202222 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4867	0.5678	0.6183	0.6222	0.6689	0.8111

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04444	0.08667	0.09349	0.12778	0.85944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04667	0.09000	0.09549	0.13111	0.66667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.056667	0.315982	0.682222	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4678	0.7050	0.7589	0.7631	0.8194	0.9922

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0033333	0.1687351	0.1626389	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1722	0.6411	0.5646	0.9394	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9628	0.9928	0.9950	0.9943	0.9967	1.0000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02611	0.05889	0.06933	0.09611	0.90056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.06111	0.07151	0.09944	0.60444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001832	0.0000000	0.8344444

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	5.054e-05	0.000e+00	4.889e-02

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001509	0.0000000	0.0527778

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002125	0.0000000	0.5483333

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9750	0.9928	0.9944	0.9943	0.9961	1.0000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01611	0.04000	0.05343	0.07333	0.87611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01681	0.04222	0.05522	0.07667	0.64056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000000 0.0000000 0.0000000 0.0000289 0.0000000 0.6377778

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.572e-06	0.000e+00	8.889e-03

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.897e-06	0.000e+00	3.889e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	4.035e-06	0.000e+00	8.333e-03

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9244	0.9917	0.9939	0.9932	0.9956	0.9989

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.03167	0.04878	0.06333	0.88111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.03333	0.04955	0.06667	0.52278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.920e-06	0.000e+00	6.556e-02

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	6.660e-07	0.000e+00	3.889e-03

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.897e-06	0.000e+00	6.111e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000399	0.0000000	0.8722222

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9378	0.9911	0.9933	0.9926	0.9950	0.9989

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.023333	0.041989	0.050000	0.897778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.010000	0.024444	0.041900	0.053333	0.527778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.00000088	0.00000000	0.3366667

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	7.560e-08	0.000e+00	1.111e-03

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8889	0.9589	0.9728	0.9694	0.9850	0.9972

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.016667	0.030745	0.037778	0.852778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.017778	0.031441	0.040000	0.456667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000277	0.0000000	0.9922222

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.002214	0.000000	1.000000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	4.86e-05	0.00e+00	1.00e+00

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9806	0.9917	0.9933	0.9932	0.9950	0.9989

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013889	0.023370	0.029444	0.698889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.014444	0.024619	0.031111	0.695556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 4.28e-05 0.00e+00 1.00e+00

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.9828 0.9917 0.9933 0.9932 0.9950 0.9994

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.011111 0.018720 0.023333 0.836667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.012222 0.019669 0.025556 0.542222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 2.430e-06 0.000e+00 1.111e-01
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      0      0      0      0      0      0

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6333	0.8428	0.8494	0.8467	0.8567	0.8756

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09111	0.13778	0.13312	0.16944	0.87722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.14111	0.13591	0.17222	0.74500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.100723	0.030556	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0766210	0.0233333	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015556 0.164932 0.123889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4944	0.6067	0.6572	0.6525	0.7022	0.8006

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04500	0.08889	0.09685	0.13167	0.83222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04667	0.09222	0.09896	0.13556	0.61611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0094444	0.1635265	0.3694444	1.0000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.875	1.000	1.000	0.995	1.000	1.000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0016667	0.0932908	0.0647222	0.9633333
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.36500	0.31859	0.50722	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9428	0.9928	0.9950	0.9944	0.9967	1.0000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02722	0.06111	0.07349	0.10000	0.84667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.02889	0.06333	0.07567	0.10333	0.68333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0001263	0.00000000	0.5505556

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0001169	0.00000000	0.0588889

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0003355	0.00000000	0.3711111

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0001083	0.00000000	0.2022222

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9378	0.9917	0.9939	0.9934	0.9956	1.0000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01722	0.04389	0.06147	0.08111	0.84611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01778	0.04556	0.06279	0.08444	0.71056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000e+00 0.000e+00 0.000e+00 1.688e-05 0.000e+00 1.439e-01
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 4.741e-06 0.000e+00 7.778e-03
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 5.669e-06 0.000e+00 1.667e-03
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00e+00 0.00e+00 0.00e+00 8.84e-05 0.00e+00 1.00e+00

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.9578 0.9922 0.9939 0.9938 0.9956 0.9994

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01222 0.03056 0.04681 0.06167 0.89167
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01333 0.03333 0.04830 0.06556 0.61778
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 9.990e-07 0.000e+00 1.667e-02
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 1.549e-07 0.000e+00 1.667e-03
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 3.543e-06 0.000e+00 5.556e-03
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 3.020e-07 0.000e+00 4.444e-03

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.5094 0.7000 0.7511 0.7423 0.7922 0.8822

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.022222	0.035896	0.045556	0.855556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.023333	0.036711	0.048889	0.502778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0009271	0.0000000	0.7466667

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3744	0.6094	0.5584	0.7722	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002462	0.0000000	0.1877778

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001878	0.000000	1.000000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9683	0.9917	0.9933	0.9934	0.9950	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007222	0.017222	0.030347	0.037222	0.857778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.018889	0.031161	0.040000	0.487778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9739	0.9917	0.9933	0.9931	0.9950	0.9989

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013889	0.025893	0.031667	0.745556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.014444	0.026777	0.033333	0.445000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.950e-06	0.000e+00	2.239e-01

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000e+00 0.000e+00 0.000e+00 9.106e-05 0.000e+00 1.428e-01
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 2.72e-07 0.00e+00 1.00e-02

```

--- Summary for K = 9 ---

Z.ar:

```

  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.9556 0.9906 0.9928 0.9923 0.9944 0.9994

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.00500 0.01278 0.02645 0.02833 0.82167
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1         1         1         1         1         1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005556 0.013333 0.026209 0.031111 0.648889
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 3.53e-05 0.00e+00 1.00e+00
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0         0         0         0         0         0
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0         0         0         0         0         0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0000247 0.0000000 0.5366667

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6867	0.8422	0.8494	0.8466	0.8561	0.8833

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.13889	0.13471	0.17056	0.86889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09722	0.14222	0.13764	0.17389	0.73389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.103601	0.032222	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0784928	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162574 0.120000 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4817	0.5861	0.6361	0.6344	0.6861	0.8039

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04667	0.09056	0.09656	0.13111	0.82222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04889	0.09333	0.09890	0.13556	0.61056
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.031667	0.251304	0.477778	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6567	0.8828	0.9111	0.9107	0.9400	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0022222	0.1578869	0.1402778	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04944	0.43778	0.43266	0.69667	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9744	0.9928	0.9950	0.9943	0.9967	1.0000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.06111	0.07309	0.09944	0.85667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03000	0.06333	0.07537	0.10333	0.67333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001219	0.0000000	0.5894444

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.937e-05	0.000e+00	3.889e-03

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001488	0.0000000	0.0388889

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001387	0.0000000	0.1977778

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9356	0.9922	0.9944	0.9938	0.9961	0.9994

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.04389	0.06079	0.08111	0.88889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01889	0.04667	0.06254	0.08444	0.65778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000000 0.0000000 0.0000000 0.0000734 0.0000000 0.7011111

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.563e-06	0.000e+00	8.333e-03

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.665e-05	0.000e+00	6.667e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.812e-05	0.000e+00	6.611e-02

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9622	0.9922	0.9939	0.9936	0.9956	1.0000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.03167	0.04734	0.06167	0.86722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.03333	0.04860	0.06556	0.54889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	5.68e-05	0.00e+00	1.00e+00

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.254e-07	0.000e+00	2.778e-03

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.189e-06	0.000e+00	2.778e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.320e-07	0.000e+00	3.889e-03

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9472	0.9917	0.9939	0.9935	0.9956	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.022778	0.037471	0.047778	0.875000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.010000	0.024444	0.038400	0.051111	0.482780

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.860e-07	0.000e+00	6.667e-03

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0000281	0.00000000	0.6377778

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9806	0.9917	0.9939	0.9935	0.9956	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.007222	0.017778	0.029905	0.038333	0.835556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.018889	0.030890	0.041111	0.428889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9828	0.9917	0.9939	0.9932	0.9950	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.014444	0.025693	0.031667	0.815556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.015556	0.026676	0.033889	0.471111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0e+00	0.0e+00	0.0e+00	3.5e-05	0.0e+00	1.0e+00

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000000 0.0000000 0.0000000 0.0000272 0.0000000 0.9994444

```

--- Summary for K = 9 ---

Z.ar:

```

  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.9767 0.9911 0.9928 0.9928 0.9950 0.9989

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.00500 0.01167 0.02285 0.02667 0.80944
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.005556 0.013333 0.023164 0.028889 0.391111
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0      0      0      0      0      0
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0      0      0      0      0      0
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000e+00 0.000e+00 0.000e+00 5.315e-06 0.000e+00 8.333e-03
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000000 0.0000000 0.0000000 0.0000238 0.0000000 0.8738889

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5089	0.7867	0.8233	0.8072	0.8439	0.8767

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0900	0.1372	0.1324	0.1700	0.8494

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09278	0.14111	0.13551	0.17333	0.75222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006111	0.011111	0.103043	0.066111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0088889	0.0744937	0.0294444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.00000 0.01333 0.10389 0.23311 0.33222 1.00000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5744	0.6428	0.6772	0.6810	0.7167	0.8378

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05000	0.09222	0.09956	0.13444	0.80778
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05222	0.09556	0.10198	0.13833	0.62556
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002778	0.199444	0.283509	0.545556	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1422	0.3628	0.3933	0.4043	0.4383	0.7167
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0033333	0.1601534	0.3325000	0.9822222
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0500	0.4422	0.3660	0.5878	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7672	0.8372	0.8739	0.8699	0.9022	0.9683

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02833	0.06444	0.07924	0.10833	0.86667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03000	0.06778	0.08130	0.11222	0.63667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.04933	0.01833	1.00000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001626	0.000000	0.262778

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.02054	0.00000	0.94333

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0207574	0.0005556	1.0000000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5117	0.5917	0.6311	0.6350	0.6761	0.7983

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.04222	0.05738	0.07667	0.87000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.04444	0.05863	0.08000	0.51556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.00000 0.00000 0.00000 0.13027 0.03611 1.00000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3639	0.4306	0.4351	0.4994	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.05849	0.00000	1.00000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4217	0.4341	0.8161	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9683	0.9917	0.9939	0.9936	0.9961	1.0000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01278	0.03222	0.04813	0.06444	0.87667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.03444	0.04942	0.06778	0.53722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.624e-07	0.000e+00	2.778e-03

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.320e-07	0.000e+00	3.889e-03

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.070e-06	0.000e+00	2.944e-02

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9517	0.9917	0.9939	0.9933	0.9956	0.9989

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.023333	0.038721	0.049444	0.880000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.010000	0.02556	0.03966	0.05222	0.57056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	3.16e-05	0.00e+00	1.00e+00

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	7.750e-08	0.000e+00	2.778e-03

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9422	0.9911	0.9933	0.9928	0.9950	0.9989

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007222	0.019722	0.037064	0.044444	0.850000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.021111	0.037658	0.047778	0.516111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000079	0.0000000	0.3355556

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.340e-06	0.000e+00	8.611e-02

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9289	0.9906	0.9928	0.9922	0.9944	0.9989

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006111	0.015556	0.033294	0.036667	0.855000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.016667	0.032530	0.040000	0.604444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000307	0.0000000	0.8511111

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9539 0.9906 0.9928 0.9923 0.9944 0.9983

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00500 0.01222 0.02537 0.02833 0.84278
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005556 0.013333 0.025364 0.030000 0.570556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 2.21e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 3.86e-05 0.00e+00 1.00e+00

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5339	0.7894	0.8311	0.8104	0.8456	0.8817

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09167	0.13833	0.13373	0.17111	0.90111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.14167	0.13696	0.17556	0.69333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.025556	0.158427	0.184444	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0094444	0.0884393	0.0351389	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.007778 0.031111 0.199776 0.260556 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5261	0.6944	0.7339	0.7229	0.7617	0.8111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04611	0.09278	0.09781	0.13389	0.83611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04778	0.09667	0.10043	0.13889	0.63722
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0283333	0.3466888	0.8361111	1.0000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0027778	0.2025574	0.1670833	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.07167	0.82167	0.61118	0.96111	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8839	0.9911	0.9939	0.9929	0.9961	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03056	0.06944	0.08765	0.11556	0.87611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.03222	0.07222	0.08907	0.12000	0.65222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001767	0.0000000	0.5055556

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001906	0.0000000	0.0594444

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001198	0.0000000	0.0316667

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001001	0.0000000	0.1977778

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9600	0.9922	0.9944	0.9938	0.9961	1.0000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01833	0.04583	0.06228	0.08278	0.82722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.02000	0.04778	0.06414	0.08667	0.75722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000e+00 0.000e+00 0.000e+00 3.162e-05 0.000e+00 2.156e-01
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 1.353e-05 0.000e+00 1.722e-02
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 1.382e-05 0.000e+00 3.889e-03
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000000 0.0000000 0.0000000 0.0001802 0.0000000 1.0000000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.4994 0.6383 0.6900 0.6869 0.7400 0.8528

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01222 0.03333 0.04799 0.06556 0.81278
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01333 0.03556 0.04962 0.06889 0.53778
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.00000 0.00000 0.01101 0.00000 1.00000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.8189 0.9144 0.8828 0.9983 1.0000
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.000000 0.000000 0.005507 0.000000 0.381111
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.0247 0.0000 0.8050

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.5128 0.6889 0.7367 0.7326 0.7817 0.8711

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.009444	0.025000	0.039897	0.052222	0.857778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.010000	0.02667	0.04089	0.05486	0.43222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0004177	0.00000000	0.6261111

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4128	0.6367	0.5873	0.7989	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0007404	0.00000000	0.6816667

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9189	0.9900	0.9928	0.9921	0.9944	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.008333	0.021111	0.041498	0.047778	0.866667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.022222	0.040849	0.051111	0.586667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9144	0.9733	0.9833	0.9799	0.9900	0.9972

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006111	0.015556	0.029677	0.035556	0.755556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.016667	0.030292	0.037778	0.552222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.820e-07	0.000e+00	2.167e-02

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001133	0.000000	1.000000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9700 0.9911 0.9928 0.9926 0.9944 0.9994

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00500 0.01278 0.02548 0.02889 0.67778
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005556 0.014444 0.026026 0.031111 0.578889
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 1.76e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0000148 0.0000000 0.5444444

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5272	0.7817	0.8206	0.8005	0.8389	0.8700

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09222	0.13889	0.13510	0.17222	0.85556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09514	0.14333	0.13848	0.17667	0.74833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.040556	0.188161	0.261111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.006111	0.010000	0.102902	0.057917	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.008889 0.033333 0.192518 0.236667 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4861	0.7383	0.7739	0.7581	0.7911	0.8250

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04278	0.09056	0.09700	0.13333	0.80000
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04444	0.09444	0.09967	0.13778	0.61278
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0338889	0.3878353	1.0000000	1.0000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0027778	0.2273090	0.2452778	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08667	1.00000	0.66809	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4956	0.6089	0.6628	0.6589	0.7111	0.8122

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02625	0.06278	0.07432	0.10389	0.77167
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.06556	0.07674	0.10778	0.55444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.1462432	0.2888889	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7839	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.07953	0.01611	0.84889

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.263333	0.263628	0.461667	1.000000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9594	0.9922	0.9939	0.9937	0.9961	0.9994

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01889	0.04722	0.06355	0.08611	0.83889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.05000	0.06553	0.09000	0.61500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.0000000 0.0000000 0.0000000 0.0000548 0.0000000 0.7266667

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001123	0.0000000	0.2500000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	8.149e-06	0.000e+00	3.889e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.000049	0.0000000	0.453333

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9317	0.9917	0.9939	0.9932	0.9956	0.9994

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01444	0.03667	0.05637	0.07222	0.85889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01556	0.03889	0.05662	0.07556	0.54778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000134	0.0000000	0.7277778

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	7.750e-08	0.000e+00	5.556e-04

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	5.315e-06	0.000e+00	8.333e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.569e-07	0.000e+00	2.778e-03

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9061	0.9844	0.9889	0.9875	0.9922	0.9978

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.02722	0.05020	0.06056	0.86833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01111	0.02889	0.04993	0.06444	0.62778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	2.28e-05	0.00e+00	1.00e+00

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0006099	0.0000000	1.0000000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.543e-07	0.000e+00	5.556e-04

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6300	0.8372	0.8778	0.8634	0.9050	0.9467

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.021111	0.039771	0.047778	0.797222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.022222	0.039539	0.051111	0.443889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1353	0.2478	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	2.340e-06	0.000e+00	8.611e-02

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9678	0.9911	0.9928	0.9926	0.9944	0.9989

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.016111	0.030763	0.036667	0.796111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.017778	0.031352	0.040000	0.652222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	1.83e-05	0.00e+00	1.00e+00

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 3.51e-05 0.00e+00 1.00e+00

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.9728 0.9906 0.9928 0.9924 0.9944 0.9989

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.013333 0.027176 0.030556 0.720000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1      1      1      1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.014444 0.027512 0.033333 0.395556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 2.72e-05 0.00e+00 1.00e+00

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6772	0.7083	0.7156	0.7152	0.7222	0.7433

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08667	0.13278	0.12748	0.16500	0.86500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08889	0.13556	0.12997	0.16778	0.75444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101928	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0772261	0.0234722	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.0005556 0.0061111 0.0150000 0.1623028 0.1183333 1.0000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5172	0.5456	0.5544	0.5543	0.5633	0.5956

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04167	0.09000	0.09145	0.12833	0.84889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04333	0.09333	0.09398	0.13222	0.66167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.210000	0.430892	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0044444	0.2398080	0.3369444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1167	1.0000	0.6903	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3689	0.3944	0.4017	0.4023	0.4100	0.4461

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02167	0.05861	0.06721	0.09764	0.82444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.06111	0.06983	0.10222	0.70111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6633	0.9672	0.7493	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2251162	0.2344444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1356	1.0000	0.6791	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7561	0.9394	0.9533	0.9522	0.9661	0.9883

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01500	0.04611	0.09511	0.09722	0.89778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.04889	0.08464	0.10111	0.69556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000000 0.0000000 0.0000000 0.0009159 0.0000000 1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	8.087e-05	0.000e+00	1.117e-01

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002987	0.0000000	0.0677778

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.004082	0.0000000	1.0000000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4156	0.4539	0.4667	0.4663	0.4789	0.5200

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.009444	0.028889	0.039629	0.057222	0.826667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01000	0.03111	0.04155	0.06111	0.55778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3080	0.7383	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6983	0.7617	0.7495	0.8139	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.09515	0.000000	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1189	0.5756	0.5335	0.9528	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8906	0.9806	0.9850	0.9840	0.9883	0.9956

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.022778	0.042130	0.050000	0.868333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.024444	0.042081	0.053333	0.518889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002271	0.0000000	1.0000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	4.650e-08	0.000e+00	1.667e-03

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	3.35e-05	0.00e+00	1.00e+00

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5494	0.6433	0.6672	0.6639	0.6867	0.7383

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006111	0.018333	0.028661	0.039444	0.733889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.020000	0.029961	0.042222	0.398333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.004753	0.000000	0.942778

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2711	0.5117	0.5210	0.7750	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001922	0.000000	0.565000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.01199	0.00000	0.70000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8417	0.9106	0.9256	0.9217	0.9367	0.9650

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00500	0.01278	0.02197	0.02833	0.73222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013333	0.023026	0.030000	0.385000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.01424	0.00000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000000 0.000000 0.000000 0.004094 0.000000 1.000000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000000 0.0000000 0.0000000 0.0009345 0.0000000 1.0000000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.7911 0.8867 0.9006 0.8987 0.9133 0.9472

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.003889 0.010556 0.018391 0.023333 0.788889
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.004444 0.011111 0.019104 0.025556 0.359444
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.000000 0.000000 0.003719 0.000000 1.000000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.000000 0.000000 0.004908 0.000000 1.000000
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.000000 0.000000 0.001913 0.000000 1.000000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.000000 0.000000 0.03183 0.000000 1.000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6850	0.7078	0.7144	0.7149	0.7222	0.7439

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09278	0.13722	0.13204	0.16778	0.84556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14000	0.13478	0.17111	0.76389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101966	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0773990	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162394 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4622	0.4967	0.5044	0.5040	0.5117	0.5367

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04778	0.09722	0.09920	0.13611	0.91000
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0500	0.1011	0.1020	0.1400	0.6922
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.058333	0.413768	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2480509	0.3904167	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1228	1.0000	0.6968	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4294	0.4550	0.4644	0.4642	0.4728	0.5067

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.05944	0.06859	0.09833	0.89111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.06222	0.07092	0.10222	0.60056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4672	0.7611	0.6738	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7706	1.0000	1.0000	0.9998	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2322215	0.3659722	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1728	1.0000	0.6859	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4444	0.4794	0.4911	0.4921	0.5039	0.5444

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01389	0.03889	0.05129	0.07222	0.82222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04111	0.05293	0.07556	0.55222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.3517 0.4479 0.8978 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.1161 0.7639 0.8044 0.8029 0.8439 1.0000
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.1744 0.1736 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.07278 0.49889 0.47684 0.83056 1.00000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.8744 0.9844 0.9878 0.9870 0.9911 0.9972

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01222 0.03111 0.05088 0.06389 0.87833
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01333 0.03333 0.05135 0.06764 0.59222
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 3.860e-06 0.000e+00 1.789e-01
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 2.479e-07 0.000e+00 2.222e-03
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 3.543e-07 0.000e+00 5.556e-04
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 1.058e-06 0.000e+00 1.278e-02

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.6400 0.7872 0.8217 0.8132 0.8467 0.8978

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.025000	0.045040	0.052778	0.875556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.010000	0.02667	0.04405	0.05611	0.49667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.04745	0.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0006463	0.00000000	0.6994444

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.0050	0.0000	0.8889

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.03708	0.000000	1.000000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8578	0.9811	0.9856	0.9845	0.9889	0.9961

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007222	0.018333	0.037290	0.041667	0.883333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.020000	0.035654	0.044444	0.475000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	5.510e-06	0.000e+00	3.128e-01

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	9.52e-05	0.00e+00	1.00e+00

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5144	0.5811	0.5972	0.5959	0.6122	0.6528

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.021111	0.032500	0.046111	0.714444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.023333	0.034481	0.050000	0.468889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.0179	0.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3550	0.4883	0.5552	0.7450	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.01075 0.00000 0.44000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.04384 0.03389 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8311  0.9139  0.9289  0.9255  0.9394  0.9633

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003889 0.010000 0.018735 0.023194 0.795000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.011111 0.019661 0.024444 0.357222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 5.29e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.02207 0.00000 1.00000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    0         0         0         0         0         0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.002031 0.000000 1.000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6700	0.7083	0.7144	0.7151	0.7217	0.7500

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.13778	0.13393	0.16944	0.93278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14111	0.13667	0.17333	0.74778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102096	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774911	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162299 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5217	0.5578	0.5661	0.5662	0.5744	0.6011

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03778	0.08944	0.09407	0.13167	0.87667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04000	0.09278	0.09644	0.13556	0.65889
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.057778	0.409424	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0033333	0.2406558	0.2993056	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09833	1.00000	0.68760	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3767	0.4089	0.4172	0.4171	0.4256	0.4844

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.06556	0.07320	0.10444	0.84444
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02667	0.06889	0.07588	0.10889	0.60333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4789	0.7033	0.6556	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.245628	0.449444	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3406	1.0000	0.7063	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4917	0.5617	0.5817	0.5813	0.6028	0.6728

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04000	0.05173	0.07222	0.69278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01556	0.04222	0.05335	0.07556	0.54722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.2411 0.3431 0.6644 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.4994 0.5378 0.5375 0.5756 0.8800
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.10599 0.01667 1.00000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.2783 0.3199 0.6161 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.6344 0.7344 0.7644 0.7607 0.7917 0.8544

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01278 0.03167 0.04803 0.06222 0.84056
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1         1         1         1         1         1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01333 0.03333 0.04882 0.06556 0.58556
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.03854 0.00000 1.00000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.05056 0.11000 0.11064 0.16111 0.76222
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.01558 0.00000 0.83722
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1329 0.2328 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.6006 0.6872 0.7144 0.7117 0.7383 0.8039

```

beta.ar:

Modality 1 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.000000	0.007222	0.021111	0.034756	0.045000	0.812778	

Modality 2 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1	1	1	1	1	1	

Modality 3 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1	1	1	1	1	1	

Modality 4 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1	1	1	1	1	1	

Modality 5 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1	1	1	1	1	1	

gamma.ar:

Modality 1 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.000000	0.007778	0.022222	0.035732	0.047778	0.524444	

Modality 2 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.0000	0.0000	0.0000	0.1253	0.0000	1.0000	

Modality 3 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.0000	0.0000	0.1039	0.1160	0.1967	0.8894	

Modality 4 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.00000	0.00000	0.00000	0.02799	0.00000	1.00000	

Modality 5 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.00000	0.00000	0.00000	0.08132	0.00000	1.00000	

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8400	0.9394	0.9533	0.9502	0.9650	0.9867

beta.ar:

Modality 1 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
0.000000	0.006667	0.020000	0.051589	0.049444	0.883889	

Modality 2 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1	1	1	1	1	1	

Modality 3 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1	1	1	1	1	1	

Modality 4 :						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1	1	1	1	1	1	

Modality 5 :						
--------------	--	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.021111	0.045080	0.052222	0.507222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0004923	0.0000000	1.0000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002974	0.0000000	1.0000000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.003635	0.000000	1.000000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9600	0.9833	0.9867	0.9864	0.9900	0.9972

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013889	0.025879	0.031667	0.861111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.015556	0.026612	0.033333	0.440000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.7528 0.8617 0.8772 0.8754 0.8917 0.9267

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.004444 0.012222 0.025532 0.028333 0.846111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.004444 0.013333 0.025062 0.031111 0.439444
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 4.58e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.07368 0.000000 1.000000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 1.025e-05 0.000e+00 2.356e-01

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6806	0.7083	0.7150	0.7155	0.7233	0.7578

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09167	0.13722	0.13268	0.16944	0.93611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.14111	0.13565	0.17333	0.75444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102127	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0775911	0.0251389	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162401 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5639	0.5978	0.6061	0.6060	0.6144	0.6417

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03944	0.08556	0.09208	0.12833	0.85000
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04111	0.08889	0.09442	0.13222	0.63500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002778	0.107222	0.414060	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0033333	0.2370376	0.2490278	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08278	1.00000	0.68297	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4272	0.4817	0.4922	0.4917	0.5017	0.5322

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02278	0.06222	0.07288	0.10333	0.85500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.06556	0.07503	0.10778	0.59611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.1600000	0.3942264	0.8738889	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9328	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2392673	0.4737500	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6522	0.9028	0.7417	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5850	0.6689	0.7028	0.6986	0.7311	0.7939

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.04278	0.05800	0.07833	0.82833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01889	0.04556	0.05936	0.08222	0.55278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1015 0.1150 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.2533 0.2906 0.2893 0.3250 0.6711
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.04755 0.00000 0.83833
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.2483 0.2601 0.4411 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.7917 0.9783 0.9839 0.9816 0.9883 0.9956

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01222 0.03556 0.07478 0.07944 0.90000
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01333 0.03778 0.06592 0.08333 0.53833
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0000124 0.0000000 0.5977778
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 1.658e-06 0.000e+00 9.444e-03
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 3.543e-07 0.000e+00 5.556e-04
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 2.740e-06 0.000e+00 6.389e-02

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.4094 0.4528 0.4633 0.4633 0.4750 0.5322

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006111 0.023333 0.037184 0.052778 0.810556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006667 0.025556 0.039081 0.056667 0.486111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.09829 0.02889 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  1.0000  1.0000  0.9597  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.05064 0.00000 0.95833
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.0000  0.2022  0.2563  0.4483  1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.4517  0.5106  0.5261  0.5250  0.5406  0.5783

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.00500 0.02111 0.03495 0.04833 0.70111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.023333	0.036854	0.052222	0.602222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.03883	0.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.5917	0.8300	0.7745	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.01887	0.000000	0.66611

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.0974	0.1622	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8444	0.9056	0.9200	0.9175	0.9317	0.9617

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013889	0.026664	0.031667	0.857778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.015556	0.027008	0.034444	0.435000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.005115	0.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000136	0.0000000	0.4305556

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```
0.000000 0.000000 0.000000 0.002411 0.000000 1.000000
```

```
Modality 5 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.000000 0.000000 0.02313 0.000000 1.000000
```

```
--- Summary for K = 9 ---
```

```
Z.ar:
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.9528 0.9833 0.9861 0.9859 0.9889 0.9961
```

```
beta.ar:
```

```
Modality 1 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.00500 0.01222 0.02351 0.02778 0.83556
```

```
Modality 2 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
```

```
Modality 3 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
```

```
Modality 4 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
```

```
Modality 5 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
```

```
gamma.ar:
```

```
Modality 1 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.005556 0.013333 0.023903 0.030000 0.615556
```

```
Modality 2 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000000 0.0000000 0.0000000 0.0000141 0.0000000 0.8022222
```

```
Modality 3 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0 0 0 0 0 0
```

```
Modality 4 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0 0 0 0 0 0
```

```
Modality 5 :
```

```
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00e+00 0.00e+00 0.00e+00 4.68e-05 0.00e+00 1.00e+00
```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6767	0.7083	0.7150	0.7151	0.7222	0.7439

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.14000	0.13492	0.17111	0.85333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14333	0.13803	0.17444	0.76667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102114	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0769746	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162265 0.119444 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4728	0.5167	0.5261	0.5259	0.5367	0.5711

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04500	0.09611	0.09905	0.13722	0.89500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04667	0.10000	0.10179	0.14111	0.63667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.101111	0.403681	0.955556	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0044444	0.2349203	0.3261111	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1133	0.9478	0.6727	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3656	0.3889	0.3972	0.3980	0.4061	0.4689

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.06667	0.07576	0.11000	0.88778
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.07000	0.07860	0.11500	0.58667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2350	0.4272	0.5399	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.264730	0.608056	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6750	1.0000	0.7863	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3106	0.3400	0.3483	0.3499	0.3594	0.4106

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01611	0.05000	0.06231	0.09000	0.79556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.05333	0.06496	0.09556	0.64111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000	0.1944	0.4944	0.5100	0.8622	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9656	1.0000	1.0000	1.0000	1.0000	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2208	0.3746	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6572	0.9017	0.7614	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3856	0.4317	0.4439	0.4432	0.4561	0.5017

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.033889	0.048377	0.071667	0.867222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.035556	0.050176	0.076667	0.608889
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1986	0.3878	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2472	1.0000	1.0000	0.9924	1.0000	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.10511	0.01194	1.000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2194	0.4278	0.4114	0.6011	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3650	0.3989	0.4111	0.4117	0.4228	0.4661

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007222 0.028889 0.042279 0.061111 0.820000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007778 0.031111 0.044770 0.066111 0.467778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1386 0.2356 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.9728 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.06892 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.3117 0.3210 0.5200 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.4289 0.4856 0.4994 0.5010 0.5167 0.5622

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006667 0.028889 0.043290 0.061111 0.784444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.031111	0.045395	0.066667	0.595556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.05974	0.05167	0.89333

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5911	0.7467	0.7499	0.9621	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.02761	0.000000	0.65500

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.06889	0.13584	0.24667	0.91611

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8528	0.9472	0.9606	0.9554	0.9689	0.9844

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.015000	0.030927	0.035000	0.857778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.016667	0.030388	0.037778	0.493333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.000883	0.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000188	0.0000000	0.4127778

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000e+00 0.000e+00 0.000e+00 4.854e-05 0.000e+00 7.611e-02
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.00518 0.00000 1.00000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.5617 0.6739 0.6894 0.6869 0.7044 0.7456

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.013333 0.027235 0.033889 0.766667
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.014444 0.027989 0.036667 0.592222
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.00145 0.00000 1.00000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0400 0.2425 0.3877 0.7511 1.0000
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0006048 0.0000000 0.3338889
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.003225 0.000000 1.000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6733	0.7078	0.7156	0.7151	0.7222	0.7517

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.13889	0.13506	0.17056	0.84667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14222	0.13848	0.17444	0.73444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101847	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0775404	0.0245833	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162741 0.120556 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5706	0.5933	0.6028	0.6025	0.6106	0.6361

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04736	0.09389	0.10047	0.13556	0.86889
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05000	0.09778	0.10310	0.14000	0.62722
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.036667	0.402309	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0027778	0.2383819	0.2645833	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08056	1.00000	0.68532	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4433	0.4828	0.4933	0.4935	0.5039	0.5372

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.06333	0.07358	0.10389	0.86111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.06556	0.07597	0.10778	0.70222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2828	0.5367	0.5626	0.9822	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5772	0.9489	0.9750	0.9687	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2163	0.2406	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3500	0.9411	0.6890	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3433	0.3689	0.3772	0.3782	0.3872	0.4678

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01611	0.05111	0.06331	0.09222	0.78667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.05444	0.06600	0.09778	0.56889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.1289 0.4239 0.4623 0.7867 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.6939 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.1907 0.2658 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.6344 0.8467 0.7362 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.7433 0.9722 0.9811 0.9779 0.9867 0.9972

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.01389 0.04056 0.09939 0.09889 0.89611
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.01444 0.04278 0.07880 0.09889 0.57333
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000e+00 0.000e+00 0.000e+00 1.744e-06 0.000e+00 1.444e-02
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000e+00 0.000e+00 0.000e+00 1.518e-06 0.000e+00 6.667e-03
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000e+00 0.000e+00 0.000e+00 3.543e-07 0.000e+00 5.556e-04
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000000 0.0000000 0.0000000 0.0000437 0.0000000 0.8355556

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.8328 0.9828 0.9867 0.9854 0.9894 0.9983

```


beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.02611	0.04746	0.05778	0.86889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.02778	0.04644	0.06111	0.46833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	9.800e-09	0.000e+00	5.556e-04

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.550e-08	0.000e+00	5.556e-04

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7678	0.8478	0.8711	0.8668	0.8900	0.9350

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007222	0.021111	0.045381	0.051111	0.841667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.023333	0.043083	0.054444	0.531667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.03329	0.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0002087	0.00000000	0.4783333

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000000	0.00000000	0.00000000	0.0007611	0.00000000	0.3955556

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.01286	0.000000	1.000000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8967	0.9806	0.9844	0.9837	0.9878	0.9961

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.006111	0.016111	0.034912	0.038889	0.839444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.006667	0.017778	0.033780	0.041111	0.453889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	5.010e-06	0.000e+00	2.839e-01

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001385 0.0000000 1.0000000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.8689 0.9800 0.9839 0.9831 0.9872 0.9956

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.00500 0.01389 0.03421 0.03444 0.85944
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.015556 0.031285 0.036667 0.496111
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0000088 0.0000000 0.4983333
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0      0      0      0      0      0
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0      0      0      0      0      0

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6872	0.7078	0.7150	0.7152	0.7228	0.7472

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09625	0.14000	0.13672	0.17167	0.87278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1000	0.1444	0.1403	0.1761	0.7622

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101971	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0773494	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162295 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4650	0.4950	0.5033	0.5034	0.5117	0.5439

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05667	0.10444	0.10723	0.14333	0.88722
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05889	0.10889	0.11037	0.14889	0.67778
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.057778	0.413477	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2472902	0.3851389	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1217	1.0000	0.6962	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3522	0.3772	0.3861	0.3858	0.3950	0.4200

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03444	0.07944	0.08526	0.11889	0.72944
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03667	0.08333	0.08866	0.12444	0.59056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5450	0.7894	0.6943	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.241661	0.378889	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3850	1.0000	0.7238	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7256	0.9739	0.9811	0.9781	0.9872	0.9972

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01833	0.05444	0.11476	0.12000	0.90444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.05667	0.09306	0.12000	0.61056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00e+00 0.00e+00 0.00e+00 4.88e-05 0.00e+00 4.15e-01
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 7.809e-06 0.000e+00 1.167e-02
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 2.905e-05 0.000e+00 1.944e-02
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000e+00 0.000e+00 0.000e+00 1.684e-05 0.000e+00 1.244e-01

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.2756 0.3206 0.3300 0.3326 0.3422 0.4400

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01389 0.04611 0.06068 0.08556 0.84444
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01444 0.04889 0.06316 0.09111 0.58389
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.1333 0.3199 0.6522 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.5861 1.0000 1.0000 0.9995 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.1769 0.2211 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.4672 0.7339 0.6383 0.8939 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.4367 0.5117 0.5289 0.5293 0.5506 0.5956

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.009444 0.035000 0.050311 0.070556 0.747222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.010000 0.03778 0.05180 0.07444 0.56056
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.00000 0.07748 0.12056 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.7144 0.8506 0.8195 0.9672 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.00000 0.03569 0.00000 0.69222
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.1694 0.1794 0.2939 0.8378

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.7133 0.8511 0.8750 0.8691 0.8911 0.9328

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.007778 0.023333 0.061498 0.061111 0.866111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :

```


Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.025000	0.051581	0.063333	0.527222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.009329	0.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001267	0.0000000	0.7994444

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001291	0.000000	0.513333

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.05248	0.000000	1.000000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9050	0.9828	0.9861	0.9855	0.9889	0.9956

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006111	0.016667	0.033121	0.038333	0.862222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.017778	0.033059	0.041111	0.473333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	3.63e-05	0.00e+00	1.00e+00

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9378  0.9822  0.9856  0.9851  0.9883  0.9961

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00500 0.01333 0.02651 0.03111 0.80722
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005556 0.014444 0.026851 0.033333 0.413333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000e+00 0.000e+00 0.000e+00 8.600e-07 0.000e+00 4.889e-02
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00e+00 0.00e+00 0.00e+00 8.52e-05 0.00e+00 1.00e+00

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5683	0.5961	0.6028	0.6037	0.6111	0.6433

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08889	0.13444	0.12941	0.16611	0.90333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09167	0.13778	0.13191	0.16889	0.77944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102044	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774713	0.0250000	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162316 0.119444 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3061	0.3311	0.3394	0.3396	0.3478	0.3828

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05000	0.09778	0.09822	0.13444	0.91167
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05222	0.10111	0.10110	0.13889	0.72944
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.059444	0.415986	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.250361	0.375556	1.000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1278	1.0000	0.6990	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1328	0.1594	0.1667	0.1679	0.1750	0.2583

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02569	0.07278	0.07623	0.11000	0.87056
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.07694	0.07980	0.11556	0.65889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6078	0.7439	0.6954	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.002778	0.253597	0.406806	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3850	1.0000	0.7474	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.08667	0.11278	0.12000	0.12129	0.12778	0.21278

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02056	0.05611	0.06334	0.09222	0.81056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.06000	0.06699	0.09833	0.59778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5656 0.8661 0.7126 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2618 0.5910 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9744 1.0000 0.8531 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.2261 0.2600 0.2728 0.2758 0.2889 0.3628

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.008333 0.027778 0.040051 0.058889 0.765556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.008889 0.030000 0.042148 0.062222 0.485556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.09444 0.31476 0.60667 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.2111 0.8367 0.8944 0.8788 0.9378 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.1327448 0.0005556 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.3617 0.8106 0.6560 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.08278 0.10889 0.12278 0.12646 0.13889 0.23889

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006667 0.033889 0.044932 0.069444 0.770000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007778 0.036667 0.048531 0.076111 0.578333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.3978 0.4234 0.7878 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.5111 1.0000 1.0000 0.9980 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1937 0.3472 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.5350 0.7578 0.6603 0.8833 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.1233 0.1528 0.1661 0.1716 0.1872 0.3489

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.00500 0.02389 0.03604 0.05444 0.74167
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.026667	0.039564	0.061111	0.482222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2123	0.4717	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2928	0.9729	1.0000	0.9692	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1032	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4494	0.4048	0.6617	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5183	0.6100	0.6256	0.6231	0.6361	0.6744

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.017778	0.028225	0.039444	0.733333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00625	0.02000	0.03017	0.04333	0.37000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.02165	0.00000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2294	0.3011	0.3434	0.4083	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.000000 0.000000 0.000000 0.008301 0.000000 0.985000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.05498 0.05333 1.00000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8928  0.9456  0.9572  0.9542  0.9656  0.9811

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.010556 0.021558 0.025000 0.805000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.011944 0.022322 0.027778 0.360000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 5.15e-05 0.00e+00 1.00e+00
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.003061 0.000000 1.000000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0001366 0.0000000 1.0000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5678	0.5972	0.6044	0.6041	0.6122	0.6422

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09278	0.13722	0.13272	0.16889	0.86667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14000	0.13546	0.17167	0.82667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101986	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0777523	0.0233333	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162441 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3778	0.4061	0.4150	0.4148	0.4233	0.4483

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04944	0.09500	0.09827	0.13278	0.84611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05222	0.09778	0.10081	0.13667	0.65833
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.045000	0.407114	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0038889	0.2420454	0.3166667	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09889	1.00000	0.69040	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2150	0.2444	0.2539	0.2557	0.2644	0.3667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.05611	0.06718	0.09778	0.86667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.05889	0.06970	0.10222	0.64333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1367	0.3278	0.4993	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.002222	0.271035	0.655000	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7817	1.0000	0.8106	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07167	0.10722	0.11500	0.11606	0.12278	0.20111

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01833	0.05778	0.06524	0.09444	0.84167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.06167	0.06882	0.10056	0.63444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5822 0.8517 0.7128 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2643 0.6035 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9439 1.0000 0.8508 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.3022 0.3344 0.3456 0.3480 0.3600 0.4406

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.009444 0.028889 0.044103 0.058889 0.847222
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01000 0.03000 0.04443 0.06222 0.56611
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1812 0.2706 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.8233 0.8828 0.8657 0.9289 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.09816 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.1856 0.7050 0.5893 0.9661 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.4911 0.5511 0.5667 0.5652 0.5806 0.6233

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007778 0.021667 0.032623 0.044444 0.806111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.008889 0.023333 0.033811 0.047778 0.513889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1181 0.1422 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.2356 0.3206 0.3104 0.3943 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.02581 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1923 0.3594 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.1161 0.1456 0.1583 0.1646 0.1789 0.3844

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.00500 0.02444 0.03818 0.05778 0.75167
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.027778	0.041698	0.064444	0.495556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2035	0.4178	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9805	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1047	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06167	0.42556	0.41031	0.67167	1.00000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2739	0.3089	0.3228	0.3254	0.3389	0.4094

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.025000	0.036561	0.053333	0.743889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.027778	0.039774	0.060000	0.510556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.0816	0.1167	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6300	0.7369	0.7554	0.9061	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.03966 0.00000 0.80056
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.1522  0.1857  0.3189  1.0000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8794  0.9661  0.9728  0.9707  0.9778  0.9883

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.011111 0.027615 0.027222 0.830000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.012222 0.026640 0.030000 0.482778
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0000127 0.0000000 0.6411111
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0004756 0.0000000 1.0000000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0001895 0.0000000 0.2972222
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0000095 0.0000000 0.3483333

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5594	0.5956	0.6033	0.6034	0.6111	0.6478

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.13778	0.13345	0.16944	0.91444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14111	0.13627	0.17278	0.75667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102081	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.077614	0.024444	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162325 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4072	0.4356	0.4422	0.4428	0.4506	0.4811

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04833	0.09278	0.09897	0.13333	0.87833
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05000	0.09556	0.10138	0.13778	0.73556
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.040000	0.403993	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0033333	0.2395904	0.2652778	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08556	1.00000	0.68719	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1983	0.2256	0.2356	0.2367	0.2467	0.3144

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.06444	0.07341	0.10833	0.87333
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.06778	0.07619	0.11333	0.63000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2933	0.4561	0.5618	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001667	0.260528	0.574583	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6389	1.0000	0.7859	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0950	0.1228	0.1322	0.1334	0.1422	0.2211

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.06222	0.06967	0.10097	0.83889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.06667	0.07313	0.10667	0.57278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5106 0.7578 0.6698 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2604 0.5713 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.8672 1.0000 0.8347 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.1172 0.1433 0.1533 0.1575 0.1678 0.2417

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.008333 0.037222 0.050955 0.076667 0.850000
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.008889 0.040556 0.053872 0.082222 0.567778
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.2978 0.4000 0.7894 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.7033 1.0000 1.0000 0.9999 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1950 0.2824 1.0000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.6794 0.8639 0.7468 0.9944 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.2361 0.2678 0.2806 0.2821 0.2944 0.4067

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.006111 0.040556 0.071358 0.086111 0.824444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.006667 0.043333 0.065928 0.092222 0.513889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.1450 0.2286 0.4311 0.9939
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.3011 0.8456 0.9444 0.9112 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.1194 0.1974 0.8939
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.3178 0.4828 0.4329 0.5878 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.1283 0.1644 0.1778 0.1823 0.1961 0.3650

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.026111 0.041186 0.059444 0.740556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.028889	0.044186	0.065556	0.527778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1724	0.3494	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.09056	0.94389	1.00000	0.95951	1.00000	1.00000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.08287	0.00000	1.00000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3767	0.3650	0.6272	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3883	0.4350	0.4472	0.4484	0.4611	0.5356

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.022222	0.037232	0.051667	0.737778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.024444	0.039157	0.057778	0.513333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.05196	0.01722	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4567	0.5367	0.5821	0.6672	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.02343 0.00000 0.56000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.1178  0.1576  0.2800  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.5183  0.5628  0.5739  0.5746  0.5856  0.6311

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.017222 0.035109 0.042222 0.776667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.018889 0.035521 0.046667 0.400000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.007503 0.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.2428  0.3456  0.3887  0.4700  1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.003142 0.000000 0.810556
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.02066 0.00000 1.00000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5683	0.5956	0.6044	0.6040	0.6128	0.6400

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09389	0.13778	0.13403	0.17056	0.92667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14111	0.13705	0.17444	0.72111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102093	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0777243	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162347 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3150	0.3433	0.3506	0.3510	0.3583	0.3872

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05111	0.10056	0.10391	0.14111	0.89111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05333	0.10444	0.10655	0.14556	0.64500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.058333	0.414824	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.248380	0.360139	1.000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1261	1.0000	0.6976	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1583	0.1889	0.1972	0.1984	0.2067	0.2767

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03278	0.07778	0.08400	0.11833	0.78833
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03444	0.08222	0.08742	0.12389	0.64000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6150	0.7728	0.7045	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001667	0.250518	0.395694	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3556	1.0000	0.7341	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1344	0.1622	0.1728	0.1757	0.1856	0.2717

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01389	0.04833	0.06150	0.09056	0.83389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.05111	0.06443	0.09556	0.55000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000	0.2783	0.5489	0.5511	0.9261	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9578	1.0000	1.0000	1.0000	1.0000	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2516	0.5196	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7800	0.9878	0.8142	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1133	0.1422	0.1539	0.1579	0.1700	0.2611

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.03667	0.05152	0.07556	0.75944
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01111	0.04000	0.05441	0.08111	0.57167
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3350	0.4171	0.7950	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7022	1.0000	1.0000	0.9999	1.0000	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1993	0.2875	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6522	0.8989	0.7514	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1022	0.1344	0.1467	0.1507	0.1633	0.3011

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.030000	0.045468	0.067222	0.783889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.033333	0.048632	0.073333	0.509444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.02778	0.30810	0.65778	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4078	1.0000	1.0000	0.9975	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1593	0.1449	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4178	0.7161	0.6088	0.8522	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1111	0.1511	0.1656	0.1701	0.1856	0.3439

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.027222	0.040445	0.058889	0.688333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.030000	0.044239	0.065556	0.493333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1703	0.2944	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9833	1.0000	0.9739	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.0851	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3372	0.3666	0.6583	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8078	0.9389	0.9461	0.9451	0.9539	0.9744

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00500	0.01556	0.04635	0.04167	0.86833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.016667	0.037275	0.043333	0.508333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002131	0.0000000	1.0000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

      0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.006318 0.000000 1.000000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.5389 0.5867 0.5978 0.5970 0.6078 0.6500

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005000 0.01944 0.03541 0.04278 0.81278
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.021667 0.035465 0.046667 0.438889
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.007413 0.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.2072 0.3033 0.3515 0.4261 1.0000
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0005134 0.0000000 0.1716667
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.006954 0.000000 1.000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5650	0.5944	0.6028	0.6028	0.6106	0.6483

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.13944	0.13578	0.17097	0.89222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09778	0.14333	0.13879	0.17444	0.75000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102014	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.005000	0.008333	0.077235	0.025000	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162209 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2989	0.3272	0.3339	0.3344	0.3417	0.3767

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0500	0.1006	0.1032	0.1411	0.8939

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05222	0.10444	0.10610	0.14556	0.64333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.062222	0.416876	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.250476	0.359306	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1322	1.0000	0.6992	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2011	0.2256	0.2344	0.2348	0.2433	0.2883

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.06333	0.07432	0.10667	0.84056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.06667	0.07717	0.11111	0.64778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7911	1.0000	0.7794	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2250567	0.2001389	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08722	1.00000	0.67865	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.08889	0.11111	0.11889	0.12012	0.12833	0.19778

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02278	0.06333	0.07207	0.10278	0.83833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.06778	0.07591	0.10889	0.60111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5494 0.8278 0.6965 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2604 0.6026 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9706 1.0000 0.8539 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.1822 0.2156 0.2267 0.2304 0.2406 0.3456

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.009444 0.037222 0.050787 0.073889 0.797222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01000 0.04000 0.05330 0.07889 0.50000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.2222 0.3676 0.7350 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.2772 0.9463 1.0000 0.9624 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.14702 0.05347 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.4283 0.7883 0.6702 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.06833 0.09778 0.11000 0.11205 0.12333 0.21833

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.030556	0.047743	0.071111	0.799444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.033333	0.050796	0.077778	0.494444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.2956	0.3858	0.7556	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6183	1.0000	1.0000	0.9980	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1883	0.3104	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6106	0.8311	0.7156	0.9706	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8039	0.9544	0.9644	0.9611	0.9711	0.9867

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.019444	0.046500	0.048333	0.861667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.021111	0.042443	0.051111	0.566111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	8.32e-05	0.00e+00	1.00e+00

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.720e-07	0.000e+00	1.333e-02

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001235	0.000000	1.000000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2617	0.3006	0.3133	0.3152	0.3289	0.3878

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00500	0.02611	0.03946	0.05722	0.75722

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.029167	0.042713	0.064444	0.452778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1676	0.3322	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6411	0.7228	0.7478	0.8511	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.03869 0.00000 0.69722
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0600  0.1725  0.3394  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8094  0.8761  0.8872  0.8856  0.8972  0.9278

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003889 0.013333 0.052391 0.038889 0.878889
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.014444 0.037041 0.040556 0.433889
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.002541 0.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.02078 0.00000 1.00000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.002312 0.000000 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.007163 0.000000 1.000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5661	0.5956	0.6033	0.6034	0.6117	0.6389

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09611	0.14000	0.13659	0.17167	0.87389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09889	0.14333	0.13998	0.17556	0.77500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101975	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0771638	0.0229167	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162359 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3500	0.3778	0.3861	0.3860	0.3944	0.4294

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05056	0.10111	0.10382	0.14167	0.89611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05278	0.10444	0.10683	0.14667	0.61111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.052222	0.410944	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0038889	0.2446882	0.3152778	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1094	1.0000	0.6934	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1317	0.1589	0.1650	0.1654	0.1722	0.1978

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03556	0.08111	0.08657	0.12056	0.83333
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03778	0.08556	0.09035	0.12667	0.67111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8839	1.0000	0.7994	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.235221	0.302361	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1194	1.0000	0.6925	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1950	0.2206	0.2294	0.2328	0.2428	0.3300

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.05167	0.06454	0.09444	0.79389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01556	0.05556	0.06724	0.09889	0.55667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000	0.3750	0.6633	0.6191	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5883	1.0000	1.0000	0.9984	1.0000	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2238	0.4412	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4611	0.7617	0.6629	0.9928	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07778	0.11389	0.12667	0.13044	0.14444	0.23778

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.04722	0.05938	0.08708	0.82833
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.05111	0.06288	0.09333	0.59333
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08889	0.41556	0.47155	0.87278	1.00000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7594	1.0000	1.0000	0.9999	1.0000	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2195	0.3976	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7200	0.9389	0.7841	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.09333	0.11722	0.12889	0.13069	0.14111	0.22778

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007222 0.036111 0.051767 0.076667 0.749444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007778 0.040000 0.055080 0.083333 0.545000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.3167 0.6878 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.4928 1.0000 1.0000 0.9922 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1722 0.1087 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.4206 0.7667 0.6410 0.9328 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.1456 0.1811 0.1956 0.1995 0.2139 0.3522

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006111 0.029444 0.043743 0.063333 0.735556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.032222	0.047231	0.070000	0.533333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1648	0.3228	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1094	0.9639	1.0000	0.9632	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.07767	0.000000	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3139	0.3311	0.5494	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1517	0.1833	0.1944	0.1989	0.2111	0.3444

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.029444	0.046823	0.067222	0.708889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.033333	0.050652	0.075556	0.505556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1599	0.3117	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2489	0.8172	0.9006	0.8889	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.07831 0.00000 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.3950  0.3747  0.6033  1.0000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.6056  0.6439  0.6567  0.6546  0.6667  0.6978

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.020000 0.059892 0.052778 0.820556
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005556 0.022222 0.048801 0.057639 0.484444
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.001733 0.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.1239  0.2244  0.2803  0.3533  1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0009031 0.0000000 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.004148 0.000000 1.000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5561	0.5956	0.6028	0.6030	0.6106	0.6356

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14000	0.13546	0.17153	0.91500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09889	0.14389	0.13907	0.17556	0.76778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102043	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0083333	0.0773817	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162358 0.119444 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3061	0.3322	0.3406	0.3400	0.3478	0.3772

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06056	0.10722	0.11132	0.14667	0.86500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06333	0.11167	0.11460	0.15222	0.70500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060556	0.416126	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0009722	0.0050000	0.2491206	0.3836111	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1278	1.0000	0.6988	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2106	0.2417	0.2500	0.2517	0.2611	0.3317

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02944	0.07556	0.08320	0.11889	0.89500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.08000	0.08664	0.12444	0.66111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1778	0.3211	0.4971	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.945	1.000	1.000	1.000	1.000	1.000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001667	0.255644	0.509583	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7722	1.0000	0.8129	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.08167	0.10722	0.11444	0.11647	0.12444	0.19778

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01833	0.05778	0.06934	0.09944	0.82833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.06222	0.07291	0.10667	0.62000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.6283 0.9167 0.7374 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.9667 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2756 0.7360 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.8406 1.0000 0.8186 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.07611 0.10778 0.11889 0.12137 0.13000 0.21500

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01056 0.04222 0.05719 0.08556 0.78000
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
1 1 1 1 1 1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01111 0.04556 0.06037 0.09222 0.53000
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.09667 0.51611 0.50136 0.87389 1.00000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.6450 1.0000 1.0000 0.9999 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2345 0.5214 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.7644 0.9711 0.8052 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.3806 0.4417 0.4533 0.4538 0.4683 0.5422

```


beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.027778	0.043639	0.059444	0.797222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.030000	0.044303	0.063333	0.439444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.10190	0.05667	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4250	0.4961	0.4817	0.5522	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.04039	0.000000	1.00000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3061	0.3578	0.6489	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4594	0.5122	0.5250	0.5242	0.5372	0.5783

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.027222	0.067124	0.066667	0.812778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.028889	0.055183	0.070000	0.571111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.04452	0.000000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3467	0.4778	0.4946	0.6389	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.01939	0.000000	1.00000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1257	0.1844	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3944	0.4356	0.4467	0.4475	0.4589	0.5333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.022778	0.041386	0.052778	0.774444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.025556	0.041244	0.057778	0.431111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.06861	0.04389	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4039	0.5400	0.5868	0.7500	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.01802 0.00000 0.82222
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.07075 0.10778 1.00000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.7500  0.9400  0.9494  0.9469  0.9567  0.9717

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003333 0.013333 0.073575 0.050000 0.869444
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003333 0.014444 0.043153 0.048889 0.518333
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.00196 0.00000 1.00000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0      0      0      0      0      0
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0      0      0      0      0      0
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0003827 0.0000000 1.0000000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4628	0.5072	0.5150	0.5149	0.5228	0.5528

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09056	0.13500	0.13059	0.16667	0.93111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.13778	0.13307	0.16889	0.74056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102004	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.077788	0.024444	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162500 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2422	0.2644	0.2722	0.2724	0.2794	0.3189

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05056	0.09722	0.09864	0.13444	0.92722
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05333	0.10111	0.10122	0.13833	0.72222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.055000	0.412593	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2460123	0.3495833	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1167	1.0000	0.6957	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.09556	0.11667	0.12278	0.12378	0.13000	0.18389

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02611	0.07000	0.07438	0.10833	0.85111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.07333	0.07766	0.11333	0.65611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8994	1.0000	0.7974	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.228217	0.251667	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08444	1.00000	0.68146	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04111	0.05778	0.06500	0.06642	0.07333	0.15111

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.05944	0.06606	0.09444	0.83611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.06333	0.06992	0.10111	0.63222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.6022 0.8928 0.7279 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2726 0.6779 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 1.0000 1.0000 0.8637 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.02000 0.03167 0.03722 0.03822 0.04278 0.12222

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01722 0.05056 0.05814 0.08500 0.78722
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01889 0.05556 0.06251 0.09222 0.62333
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.4956 0.8139 0.6907 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2793 0.7372 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 1.0000 1.0000 0.8845 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.01556 0.03667 0.04500 0.04697 0.05333 0.14389

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.009444 0.036667 0.046485 0.068889 0.761111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.010000 0.040000 0.05031 0.07556 0.57000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.3483 0.4492 0.9489 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.7328 1.0000 1.0000 0.9999 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2248 0.4075 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.6883 0.9811 0.7757 1.0000 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.02000 0.03778 0.04611 0.04866 0.05611 0.12556

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006111 0.031111 0.042408 0.063889 0.742222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```


Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.034444	0.046656	0.071111	0.527778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3528	0.4323	0.8461	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6161	1.0000	1.0000	0.9996	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1969	0.2993	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6267	0.8794	0.7340	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6472	0.6956	0.7061	0.7052	0.7150	0.7428

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.012778	0.027411	0.032222	0.797778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.013889	0.027291	0.034444	0.383333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.04369	0.00000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.02979	0.03167	0.77556

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000000 0.000000 0.000000 0.008159 0.000000 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.0287  0.0000  1.0000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.07500 0.09889 0.11167 0.11730 0.13000 0.23889

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003333 0.016111 0.028283 0.042778 0.702222
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.018889 0.032380 0.050000 0.468889
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1093  0.1133  1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.8339  0.9411  0.9023  1.0000  1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.05186 0.00000 0.99111
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.1650  0.2498  0.4706  1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4794	0.5056	0.5133	0.5134	0.5211	0.5528

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09222	0.13611	0.13278	0.16778	0.94722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09514	0.13944	0.13537	0.17111	0.76444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102021	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0083333	0.0777122	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162376 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2067	0.2328	0.2394	0.2394	0.2467	0.2706

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05056	0.09778	0.10121	0.13667	0.91556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05222	0.10111	0.10390	0.14111	0.69167

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.063333	0.417019	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005556	0.250816	0.410000	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1350	1.0000	0.6995	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1156	0.1400	0.1467	0.1473	0.1539	0.2189

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.05722	0.06831	0.10111	0.82444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.06056	0.07114	0.10611	0.63722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5089	0.6950	0.6648	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.247309	0.404583	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4167	1.0000	0.7446	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04778	0.06667	0.07500	0.07729	0.08444	0.15333

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.06000	0.06688	0.09667	0.78667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02167	0.06444	0.07046	0.10278	0.62278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5178 0.7833 0.6855 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2642 0.5733 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9906 1.0000 0.8593 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03889 0.06000 0.06778 0.07039 0.07722 0.14889

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.008889 0.037778 0.050810 0.075556 0.841111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01000 0.04111 0.05383 0.08111 0.60500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02056 0.45889 0.50293 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.7033 1.0000 1.0000 1.0000 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2376 0.4042 1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.8478 1.0000 0.8234 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.08611 0.11556 0.12444 0.12818 0.13889 0.20444

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00500	0.02444	0.03895	0.05833	0.77389

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.026667	0.041498	0.063333	0.568889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0100	0.3145	0.7006	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3367	0.9161	0.9989	0.9514	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1253	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3061	0.6850	0.5852	0.8783	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.05444	0.07889	0.09000	0.09478	0.10667	0.23222

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.019444	0.035712	0.051667	0.747222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.022222	0.038660	0.057778	0.517222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2307	0.4794	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01278	1.00000	1.00000	0.99206	1.00000	1.00000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1042	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.48278	0.45397	0.76333	1.00000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03111	0.05000	0.06056	0.06321	0.07444	0.15111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.023889	0.037938	0.056667	0.721111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.027778	0.042244	0.064444	0.502778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2231	0.4950	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2361	1.0000	1.0000	0.9857	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.0000 0.0000 0.0000 0.1138 0.0000 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.04667 0.62222 0.51427 0.82000 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.4261 0.4672 0.4772 0.4781 0.4872 0.5583

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.015556 0.025222 0.035000 0.675000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.017778 0.027547 0.038889 0.398889
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.00881 0.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.2739 0.3683 0.4090 0.4906 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.005389 0.000000 0.660000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.04245 0.00000 1.00000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4789	0.5061	0.5133	0.5140	0.5222	0.5478

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09181	0.13833	0.13413	0.17000	0.90389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.14167	0.13688	0.17333	0.73333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102140	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774029	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162452 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1978	0.2322	0.2400	0.2401	0.2483	0.2711

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05514	0.10111	0.10470	0.14000	0.84889
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05778	0.10444	0.10771	0.14444	0.78444
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.062778	0.417010	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005556	0.250499	0.391806	1.000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1300	1.0000	0.6995	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07333	0.09000	0.09556	0.09568	0.10111	0.12389

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.07833	0.08321	0.11667	0.86556
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03333	0.08222	0.08667	0.12222	0.66333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9561	1.0000	0.8055	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.235517	0.283333	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1111	1.0000	0.6918	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04500	0.07556	0.08389	0.08477	0.09222	0.14722

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01722	0.05389	0.06421	0.09278	0.81556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01889	0.05778	0.06751	0.09889	0.62000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.4711 0.8117 0.6821 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2634 0.5608 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.9817 1.0000 0.8506 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.04000 0.05889 0.06778 0.07001 0.07778 0.15500

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01278 0.04667 0.05792 0.08611 0.80111
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01333 0.05056 0.06166 0.09222 0.52278
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.1350 0.5011 0.5135 0.9439 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.8672 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2399 0.4710 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.7928 1.0000 0.8234 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.01889 0.03833 0.04556 0.04781 0.05444 0.13111

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007222	0.035556	0.049709	0.073889	0.799444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.038889	0.053253	0.081111	0.601111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4811	0.4785	0.9144	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7822	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2362	0.5436	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7111	0.9272	0.7760	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4472	0.4822	0.4933	0.4932	0.5050	0.5783

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.023889	0.071287	0.060000	0.847778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.026667	0.053809	0.063333	0.529444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.04642	0.000000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3089	0.3822	0.3692	0.4439	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.01945	0.000000	1.00000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1228	0.2196	0.3828	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3678	0.4106	0.4222	0.4231	0.4328	0.5217

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.016111	0.032292	0.037222	0.804444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.017778	0.031510	0.041111	0.633333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1059	0.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2328	0.3383	0.3213	0.4250	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.02751 0.00000 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1585  0.2700  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.4061  0.4439  0.4544  0.4558  0.4661  0.5194

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003333 0.014444 0.028121 0.035556 0.759444
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.015556 0.029681 0.040000 0.504444
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.02666 0.00000 1.00000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.3400  0.4244  0.4353  0.5072  1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.01369 0.00000 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1066  0.1728  1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4678	0.5061	0.5139	0.5139	0.5217	0.5494

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0950	0.1394	0.1359	0.1711	0.8944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09778	0.14222	0.13881	0.17444	0.72889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102034	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0780456	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162453 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2128	0.2356	0.2428	0.2427	0.2506	0.2733

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05278	0.10278	0.10558	0.14222	0.80556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05556	0.10667	0.10858	0.14667	0.66611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060556	0.416370	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.250076	0.386250	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1328	1.0000	0.6989	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1217	0.1522	0.1606	0.1621	0.1706	0.2417

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01944	0.05833	0.06923	0.10222	0.89278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.06111	0.07181	0.10667	0.62556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7661	1.0000	0.7733	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2259396	0.2123611	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1172	1.0000	0.6837	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03833	0.06111	0.06833	0.06936	0.07556	0.14000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01944	0.06000	0.06899	0.10000	0.84833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.06444	0.07269	0.10667	0.65500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5683 0.8311 0.7057 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2704 0.6675 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9950 1.0000 0.8584 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03111 0.04944 0.05611 0.05862 0.06556 0.14222

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01556 0.04944 0.06017 0.08778 0.83556
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01667 0.05333 0.06408 0.09444 0.59889
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2744 0.6322 0.5835 0.9906 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.8622 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2524 0.5406 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.8672 1.0000 0.8440 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03667 0.05944 0.06889 0.07106 0.07944 0.15111

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.028889 0.045548 0.067222 0.799444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1          1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1          1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1          1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1          1      1      1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.031111 0.048393 0.073333 0.595556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.08778 0.36773 0.81778 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.6656  1.0000  1.0000  0.9996  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.1860  0.2053  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.4906  0.8444  0.6852  0.9828  1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.2239  0.2517  0.2622  0.2641  0.2744  0.3517

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005000 0.01778 0.03216 0.04222 0.74000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1          1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1          1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
      1      1      1          1      1      1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.020000	0.033246	0.045556	0.497778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.13399	0.08167	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5617	0.6628	0.6430	0.7489	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.04486	0.000000	1.00000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.2478	0.3440	0.6344	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7533	0.8372	0.8478	0.8463	0.8578	0.8911

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.015556	0.072820	0.047778	0.885556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.016667	0.044890	0.047778	0.505000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.01419	0.000000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0000639	0.0000000	0.7433333

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.000000 0.000000 0.000000 0.001544 0.000000 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.01349 0.00000 1.00000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.1244  0.1506  0.1611  0.1654  0.1767  0.2667

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003333 0.017778 0.030487 0.044444 0.677778
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.020000 0.034290 0.051111 0.423333
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.04228 0.00000 1.00000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.7072  0.8256  0.8108  0.9594  1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.02219 0.00000 0.86000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.09190 0.07333 1.00000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4778	0.5056	0.5139	0.5137	0.5217	0.5539

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.13889	0.13543	0.17056	0.89889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09889	0.14222	0.13846	0.17333	0.74667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102149	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774384	0.0250000	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162505 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2644	0.2917	0.2994	0.2997	0.3078	0.3328

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05278	0.09889	0.10387	0.13833	0.89222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0550	0.1022	0.1066	0.1422	0.6544
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.047222	0.408731	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0038889	0.2438644	0.3336111	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1033	1.0000	0.6921	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.08667	0.10944	0.11611	0.11663	0.12278	0.15889

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02389	0.07556	0.08046	0.11611	0.85111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.08000	0.08384	0.12222	0.63833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9278	1.0000	0.8005	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.226763	0.216528	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0900	1.0000	0.6819	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1094	0.1417	0.1522	0.1558	0.1661	0.2906

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01167	0.04778	0.05943	0.08833	0.69722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.05111	0.06214	0.09333	0.52556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.2861 0.6167 0.5786 0.9739 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.6783 1.0000 1.0000 0.9976 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2152 0.3846 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.6544 0.8983 0.7671 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.03667 0.06000 0.06778 0.07272 0.08389 0.17944

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00500 0.03333 0.05022 0.07778 0.77056
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1      1      1      1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.036667 0.053324 0.084444 0.536111
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.07556 0.47889 0.49108 0.89000 1.00000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.7717 1.0000 1.0000 0.9998 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2238 0.4246 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.8350 1.0000 0.8326 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.03222 0.05222 0.06278 0.06425 0.07389 0.14889

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.036111	0.051145	0.075000	0.782778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.040000	0.054866	0.082222	0.585556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1861	0.3854	0.8278	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6611	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1914	0.2178	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5656	0.8850	0.7218	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2717	0.3022	0.3128	0.3163	0.3283	0.4189

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.023889	0.049712	0.055556	0.798333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.025556	0.045426	0.060000	0.497778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.08475	0.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.5200	0.6006	0.5910	0.6750	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.04007	0.000000	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.3261	0.3563	0.6056	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2633	0.2944	0.3044	0.3070	0.3178	0.4172

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.00500	0.01556	0.02921	0.03556	0.75278

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.016667	0.030023	0.038889	0.487778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.12840	0.02611	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.4028	0.5056	0.4861	0.5922	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.02798 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2386  0.4544  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0800  0.1039  0.1189  0.1225  0.1350  0.2489

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.002778 0.016667 0.034590 0.050556 0.708889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.018889 0.038569 0.058889 0.450000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1769  0.3789  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.1150  0.8383  0.9183  0.9040  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.07856 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.3767  0.3454  0.5667  1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4689	0.5056	0.5139	0.5134	0.5217	0.5506

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14111	0.13688	0.17278	0.87944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09833	0.14500	0.14030	0.17667	0.75778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102139	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0083333	0.0774919	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162549 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2428	0.2661	0.2744	0.2739	0.2817	0.3056

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05278	0.10278	0.10560	0.14222	0.89278
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05556	0.10667	0.10855	0.14667	0.71611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.053889	0.412369	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2468052	0.3816667	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1161	1.0000	0.6953	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07944	0.10722	0.11500	0.11577	0.12278	0.17889

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03014	0.07889	0.08444	0.12000	0.89111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03222	0.08333	0.08798	0.12556	0.58222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7322	0.8939	0.7503	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001667	0.241536	0.343194	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2417	1.0000	0.7165	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07611	0.09833	0.10778	0.11070	0.12000	0.19889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01833	0.05611	0.06764	0.09722	0.83278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.05944	0.07069	0.10333	0.55444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.3500 0.7506 0.6412 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2618 0.5897 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.8717 1.0000 0.8252 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.04000 0.05556 0.06333 0.06747 0.07611 0.14111

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.008889 0.041111 0.055889 0.084444 0.818333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01000 0.04444 0.05931 0.09111 0.56944
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1511 0.5733 0.5463 0.9872 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.7228 1.0000 1.0000 1.0000 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2548 0.5758 1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.8206 1.0000 0.8275 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03889 0.06278 0.07611 0.07843 0.09167 0.20278

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.025556 0.044439 0.067222 0.787778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.028056 0.047834 0.073333 0.527222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.2622 0.3902 0.7867 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.6494 1.0000 1.0000 0.9995 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1906 0.2293 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.4656 0.8144 0.6740 0.9839 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.2444 0.2761 0.2850 0.2882 0.2961 0.4461

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.002778 0.021667 0.047136 0.058889 0.775556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.023333	0.044549	0.062222	0.433889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.03389	0.25092	0.47389	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4850	0.5583	0.5470	0.6261	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.0313	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0200	0.2517	0.4906	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03111	0.05111	0.06167	0.06478	0.07444	0.15333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.029444	0.046039	0.069444	0.713889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.034444	0.050980	0.079444	0.453333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.05667	0.28464	0.59056	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4772	1.0000	1.0000	0.9929	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1327 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.2983 0.6094 0.5431 0.8172 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.3978 0.4294 0.4406 0.4411 0.4522 0.5294

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.003889 0.025556 0.051440 0.060556 0.729444
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.028889 0.050347 0.067778 0.436111
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.1006 0.1928 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.4033 0.4650 0.4974 0.5467 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.01524 0.00000 0.50444
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.08178 0.15500 0.73833

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4761	0.5061	0.5139	0.5139	0.5217	0.5550

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14056	0.13673	0.17167	0.85278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09889	0.14444	0.14041	0.17556	0.70389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102020	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0773902	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162522 0.119444 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2328	0.2650	0.2728	0.2725	0.2800	0.3161

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0550	0.1033	0.1063	0.1428	0.8539

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05778	0.10722	0.10952	0.14778	0.65222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.055000	0.412614	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0044444	0.2464877	0.3484722	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1206	1.0000	0.6956	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.08556	0.10444	0.11111	0.11193	0.11833	0.18556

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03444	0.08278	0.08678	0.12222	0.84167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03667	0.08722	0.09076	0.12889	0.59333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6128	0.7583	0.7007	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.002222	0.249646	0.387361	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3678	1.0000	0.7435	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04667	0.07556	0.08389	0.08649	0.09500	0.16167

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.04944	0.06354	0.09389	0.79500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.05333	0.06667	0.10000	0.61444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.0000 0.5472 0.8878 0.7127 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2626 0.6354 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9861 1.0000 0.8488 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.03000 0.04444 0.05111 0.05342 0.06056 0.11667

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.008889 0.045000 0.057853 0.087222 0.771667
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1          1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01000 0.04889 0.06159 0.09444 0.62222
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.5494 0.8878 0.7136 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.6900 1.0000 1.0000 0.9993 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2226 0.3936 1.0000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.6917 1.0000 0.7795 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.03889 0.06056 0.07222 0.07364 0.08389 0.16111

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006667 0.031667 0.048751 0.072222 0.722222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007778 0.034444 0.052239 0.078889 0.555556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.3348 0.7361 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.6711 1.0000 1.0000 0.9997 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.16860 0.03403 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.5406 0.8306 0.6913 0.9856 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.09444 0.12111 0.13167 0.13634 0.14833 0.25556

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.003889 0.025556 0.044219 0.066667 0.755556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.027778	0.047390	0.073333	0.485556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2354	0.4917	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3683	0.9017	1.0000	0.9441	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.117796	0.009861	0.962778

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2683	0.5056	0.4557	0.6650	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07444	0.09944	0.11167	0.11503	0.12611	0.21722

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.027222	0.043638	0.064444	0.703889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.031111	0.048010	0.073333	0.449444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1874	0.3778	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1672	0.8750	0.9439	0.9253	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.09382 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.03444 0.44722 0.41492 0.68722 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.2250  0.2567  0.2672  0.2700  0.2806  0.3533

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003889 0.016111 0.033044 0.042778 0.711111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.018889 0.034984 0.048889 0.473333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.08049 0.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.4761  0.6172  0.6087  0.7450  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.01971 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1120  0.1061  1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4089	0.4350	0.4428	0.4425	0.4511	0.4794

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09111	0.13444	0.13070	0.16667	0.89667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.13778	0.13323	0.16944	0.79833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102017	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0775255	0.0240278	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162485 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1583	0.1839	0.1911	0.1914	0.1983	0.2228

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04778	0.09611	0.09797	0.13444	0.90111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0500	0.1000	0.1006	0.1389	0.6800
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.057778	0.414611	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2488552	0.3463889	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1233	1.0000	0.6974	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04833	0.07111	0.07722	0.07806	0.08444	0.13111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.07389	0.07827	0.11167	0.90222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03333	0.07778	0.08166	0.11667	0.61222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5650	0.7033	0.6794	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.002222	0.255317	0.431944	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4156	1.0000	0.7528	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01722	0.02944	0.03444	0.03493	0.03833	0.09722

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.06111	0.06668	0.09444	0.76667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02500	0.06556	0.07076	0.10111	0.61278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.7044 0.9494 0.7583 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1      1      1          1      1      1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2793 0.8004 1.0000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 1.0000 1.0000 0.8672 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.008889 0.024444 0.030000 0.031443 0.037222 0.105556

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.009444 0.038333 0.049916 0.076111 0.830556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1      1      1          1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1      1      1          1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1      1      1          1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1      1      1          1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.010000 0.042222 0.05349 0.08222 0.55333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.3744 0.7767 0.6515 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1      1      1          1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2699 0.6929 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 1.0000 1.0000 0.8748 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.004444 0.015556 0.020000 0.021080 0.025000 0.085556

```



```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.008333 0.035556 0.046442 0.070000 0.766667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.008889 0.039167 0.050373 0.077778 0.485000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.022222 0.630000 0.54810 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.8339  1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.2714  0.7210  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.8894  1.0000  0.8448  1.0000  1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.009444 0.022778 0.029444 0.030751 0.036667 0.074444

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.028333 0.043020 0.063333 0.756667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.031111	0.046923	0.071111	0.557222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3422	0.8189	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6206	1.0000	1.0000	0.9998	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.1744491	0.0004167	1.0000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5711	0.9283	0.7367	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01778	0.03389	0.04056	0.04282	0.05000	0.12444

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002222	0.015556	0.030921	0.047222	0.722778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002222	0.017778	0.034636	0.054444	0.491667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2183	0.4806	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.06611	1.00000	1.00000	0.99170	1.00000	1.00000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1027 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.5156 0.4515 0.7861 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.3711 0.3933 0.4017 0.4024 0.4106 0.4728

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003333 0.017778 0.036437 0.046111 0.762222
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.020000 0.037400 0.052222 0.423889
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.05161 0.00000 1.00000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.4128 0.4661 0.4767 0.5228 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.02187 0.00000 0.95444
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.1483 0.1915 0.3511 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4022	0.4356	0.4428	0.4431	0.4506	0.4806

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09389	0.13778	0.13356	0.16944	0.90056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14111	0.13621	0.17222	0.75667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102076	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774486	0.0250000	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162401 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1494	0.1689	0.1756	0.1754	0.1828	0.2033

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05556	0.10000	0.10306	0.13833	0.89833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05778	0.10333	0.10615	0.14222	0.67667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.063889	0.417264	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005556	0.250587	0.392778	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1333	1.0000	0.6998	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.05833	0.07611	0.08167	0.08282	0.08833	0.14056

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02889	0.07389	0.07895	0.11222	0.88556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.07778	0.08230	0.11778	0.65722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6294	0.7867	0.7092	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001667	0.248887	0.371111	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3378	1.0000	0.7358	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01611	0.03056	0.03500	0.03550	0.03944	0.08444

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01944	0.05944	0.06665	0.09556	0.78667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.06333	0.07053	0.10222	0.59556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.000	0.745	1.000	0.776	1.000	1.000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2758	0.7924	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8595	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01278	0.02722	0.03278	0.03411	0.03944	0.10667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01167	0.04333	0.05397	0.08056	0.80000
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.04667	0.05782	0.08667	0.56056
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3256	0.7483	0.6364	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2650	0.6306	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8732	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.007778	0.016667	0.020556	0.022344	0.026111	0.093889

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.008333 0.037222 0.049156 0.072222 0.787222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.008889 0.041111 0.053133 0.080000 0.595556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.08667 0.64333 0.56266 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.9672  1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0000  0.2670  0.6708  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.8906  1.0000  0.8447  1.0000  1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.01333 0.03167 0.03833 0.04023 0.04667 0.09833

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.003333 0.018889 0.034759 0.053333 0.748333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :

```


Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.021111	0.038323	0.060000	0.492222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2913	0.6883	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4672	1.0000	1.0000	0.9984	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1504	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05333	0.69056	0.56164	0.92000	1.00000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1133	0.1333	0.1428	0.1448	0.1533	0.2133

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.012778	0.026262	0.036667	0.727778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.014444	0.028796	0.041111	0.418889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1679	0.2472	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6394	0.7444	0.7227	0.8317	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.05107 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.09722 0.29323 0.59389 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.04278 0.06278 0.07278 0.07627 0.08611 0.13889

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.002778 0.017222 0.029689 0.043889 0.677778
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003333 0.020000 0.033885 0.051111 0.430000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1738 0.3228 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8333 0.9289 0.8960 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.04219 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2028 0.4072 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4056	0.4361	0.4439	0.4433	0.4517	0.4744

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09278	0.13778	0.13383	0.16944	0.88000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14111	0.13658	0.17333	0.71556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102119	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0770408	0.0233333	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162429 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1428	0.1700	0.1772	0.1770	0.1844	0.2056

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05222	0.09944	0.10362	0.13944	0.91278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05444	0.10333	0.10656	0.14333	0.65833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.062778	0.417062	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.249839	0.389028	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1317	1.0000	0.6997	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.06500	0.08556	0.09167	0.09195	0.09778	0.12889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.07167	0.07925	0.11000	0.88833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03333	0.07556	0.08245	0.11556	0.64611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8433	1.0000	0.7889	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2275120	0.2125000	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08722	1.00000	0.68184	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02167	0.03667	0.04389	0.04475	0.05222	0.07778

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01389	0.05278	0.06312	0.09333	0.84889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01556	0.05667	0.06649	0.10000	0.61111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.7561 1.0000 0.7801 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2641 0.6333 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9906 1.0000 0.8388 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.02056 0.03667 0.04556 0.04714 0.05444 0.11056

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01056 0.04167 0.05571 0.08111 0.80278
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01222 0.04556 0.05908 0.08778 0.66222
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.05944 0.50944 0.51755 1.00000 1.00000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.8778 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2441 0.4890 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9167 1.0000 0.8454 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.01111 0.02278 0.02889 0.03056 0.03611 0.08111

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.009444 0.037778 0.050960 0.075000 0.719444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.01056 0.04111 0.05500 0.08222 0.51556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.4411 0.4829 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.8378 1.0000 1.0000 1.0000 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2350 0.4422 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.7572 1.0000 0.8015 1.0000 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.005556 0.016111 0.020556 0.021932 0.026111 0.049444

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.003333 0.020556 0.037829 0.056111 0.748333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.023333	0.041388	0.063333	0.550000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3219	0.7950	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3683	1.0000	1.0000	0.9993	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1678	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6078	0.9956	0.7515	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02944	0.04778	0.05889	0.06238	0.07444	0.17000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002778	0.016111	0.033039	0.048333	0.729444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.018889	0.036732	0.055556	0.508889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2016	0.4378	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9717	1.0000	0.9743	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.0000 0.0000 0.0000 0.1008 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.4656 0.4156 0.7194 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.2861 0.3189 0.3283 0.3294 0.3389 0.3839

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003889 0.012778 0.026913 0.032222 0.730556
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.014444 0.027320 0.035556 0.448889
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.09557 0.00000 1.00000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.2500 0.3828 0.3585 0.4878 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.02563 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1342 0.1639 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4067	0.4350	0.4428	0.4426	0.4506	0.4889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0950	0.1383	0.1351	0.1700	0.8622

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09778	0.14111	0.13816	0.17389	0.74167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102045	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0772991	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162327 0.117222 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2328	0.2594	0.2678	0.2677	0.2756	0.3133

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04833	0.09111	0.09850	0.13278	0.90500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05000	0.09444	0.10090	0.13667	0.64500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.037778	0.402853	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0033333	0.2383036	0.2834722	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0850	1.0000	0.6857	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.06111	0.08333	0.08944	0.09045	0.09667	0.15667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.07278	0.08001	0.11278	0.88778
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03000	0.07667	0.08331	0.11889	0.70333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7833	0.9833	0.7740	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.234122	0.288194	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1539	1.0000	0.6964	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01778	0.03056	0.03556	0.03612	0.04167	0.07056

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02500	0.06444	0.07320	0.10278	0.82722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02667	0.06889	0.07714	0.10889	0.59333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000	0.7817	1.0000	0.7863	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2709	0.6583	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8474	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01667	0.03333	0.04111	0.04340	0.05167	0.12222

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01278	0.04444	0.05642	0.08167	0.81000
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04778	0.05973	0.08778	0.51500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1283	0.5989	0.5586	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9289	1.0000	1.0000	1.0000	1.0000	1.0000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2555	0.5671	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9533	1.0000	0.8514	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01111	0.02556	0.03111	0.03249	0.03778	0.10222

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005694 0.032222 0.047851 0.072778 0.785556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1          1          1          1          1          1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1          1          1          1          1          1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1          1          1          1          1          1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1          1          1          1          1          1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.006667 0.035556 0.051555 0.080000 0.627778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.5478 0.5232 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.7467 1.0000 1.0000 1.0000 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2468 0.5479 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000 0.7194 0.9883 0.7865 1.0000 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.01222 0.03000 0.03889 0.04004 0.04722 0.13722

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.028889 0.044537 0.067222 0.777778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1          1          1          1          1          1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1          1          1          1          1          1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1          1          1          1          1          1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.032222	0.048607	0.075556	0.554444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1833	0.3713	0.7750	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3439	1.0000	1.0000	0.9991	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1845	0.2303	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5478	0.8144	0.6908	0.9622	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01222	0.02722	0.03389	0.03570	0.04222	0.09333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00500	0.02889	0.04406	0.06611	0.71722

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.033333	0.049014	0.075556	0.502778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1217	0.3442	0.7406	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5500	1.0000	1.0000	0.9987	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.17170 0.04347 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.4322  0.7711  0.6420  0.9217  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.1028  0.1256  0.1339  0.1361  0.1450  0.1883

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.002222 0.016111 0.031311 0.046667 0.694444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.018889 0.035145 0.053333 0.445556
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1238  0.1828  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.7489  0.8150  0.8171  0.8867  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.06414 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.3522  0.3367  0.5844  1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4028	0.4344	0.4428	0.4427	0.4511	0.4839

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0950	0.1394	0.1365	0.1711	0.8717

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09778	0.14333	0.13953	0.17444	0.72667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102016	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774908	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162265 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1550	0.1794	0.1867	0.1866	0.1933	0.2178

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05389	0.10278	0.10599	0.14222	0.88722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05667	0.10667	0.10886	0.14667	0.64889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060556	0.415862	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.248209	0.352083	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1283	1.0000	0.6984	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.05722	0.07833	0.08556	0.08620	0.09278	0.15500

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03014	0.07556	0.08238	0.11444	0.81833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03222	0.08000	0.08566	0.12000	0.64333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7956	0.9933	0.7774	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.232965	0.260972	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1494	1.0000	0.6961	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02500	0.04056	0.04722	0.04941	0.05556	0.10333

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01167	0.04833	0.06141	0.09167	0.79833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.05222	0.06457	0.09778	0.59556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000	0.7078	1.0000	0.7701	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2650	0.6467	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8457	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01722	0.03111	0.03722	0.03868	0.04500	0.10611

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.03944	0.05453	0.08056	0.74389
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01111	0.04333	0.05807	0.08667	0.55833
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2667	0.7356	0.6196	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.875	1.000	1.000	1.000	1.000	1.000
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2554	0.5476	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9917	1.0000	0.8655	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03444	0.05444	0.06444	0.06762	0.07778	0.14000

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.002778 0.019444 0.037818 0.056111 0.752778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.021111 0.040650 0.061111 0.581111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.4578 0.4592 0.8900 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.1250 0.9800 1.0000 0.9809 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.168822 0.004028 1.000000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.1311 0.6678 0.5589 0.9083 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.009444 0.025000 0.030556 0.031880 0.037222 0.105556

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005556 0.031111 0.047388 0.071111 0.750000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005694	0.035556	0.051740	0.078889	0.573333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1522	0.3915	0.8633	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6072	1.0000	1.0000	0.9998	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2018	0.2508	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5544	0.8783	0.7173	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02056	0.04111	0.04944	0.05222	0.06056	0.11278

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.028333	0.045264	0.065000	0.721111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.032222	0.049295	0.073333	0.603889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2395	0.5189	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3717	1.0000	1.0000	0.9913	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1235 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.05056 0.50944 0.46019 0.75444 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.06056 0.08056 0.09111 0.09373 0.10389 0.20667

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003333 0.020556 0.036478 0.052778 0.696111
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003333 0.024444 0.040791 0.061111 0.488889
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1185 0.1367 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.8256 0.8978 0.8841 0.9689 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.05793 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.2144 0.2752 0.5033 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4056	0.4356	0.4439	0.4433	0.4517	0.4789

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09611	0.14056	0.13653	0.17167	0.92389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09889	0.14444	0.13993	0.17597	0.72000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101977	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.077962	0.023472	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162307 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1478	0.1678	0.1744	0.1749	0.1811	0.2083

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05944	0.10611	0.10999	0.14500	0.85722
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06222	0.11000	0.11342	0.15000	0.67667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.063333	0.417310	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.250914	0.395694	1.000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1372	1.0000	0.7001	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.05389	0.07333	0.07889	0.07974	0.08611	0.11889

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03444	0.07889	0.08516	0.11833	0.76889
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03667	0.08333	0.08873	0.12444	0.58444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8522	1.0000	0.7916	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.234759	0.293194	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1183	1.0000	0.6904	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02389	0.03833	0.04389	0.04566	0.05278	0.10222

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02056	0.05889	0.07017	0.10056	0.80833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.06333	0.07382	0.10667	0.61611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000	0.6783	1.0000	0.7603	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2613	0.6161	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	1.000	1.000	0.857	1.000	1.000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01333	0.02556	0.03056	0.03194	0.03667	0.11111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.04500	0.05857	0.08722	0.79444
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01111	0.04889	0.06235	0.09444	0.61000
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2850	0.7406	0.6259	1.0000	1.0000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2715	0.7111	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8728	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01500	0.03167	0.04000	0.04240	0.05222	0.08611

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.002778 0.020000 0.041093 0.062222 0.748889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.003333 0.022222 0.043763 0.067778 0.493333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.09722 0.40594 0.96556 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.3550  1.0000  1.0000  0.9978  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.1726  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.0000  0.3689  0.8600  0.6707  1.0000  1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.007222 0.020000 0.026111 0.028302 0.033889 0.104444

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.007222 0.037778 0.051915 0.077778 0.740000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.042222	0.056804	0.086667	0.508889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3794	0.4712	0.9572	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9011	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2202	0.3951	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6794	0.9717	0.7740	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02500	0.04222	0.05056	0.05258	0.06056	0.11667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.025000	0.042549	0.062778	0.702778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.028889	0.047081	0.072222	0.509444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.04278	0.32427	0.71611	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2567	0.9517	1.0000	0.9694	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1261 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.1106 0.5856 0.4997 0.8011 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.1339 0.1533 0.1633 0.1652 0.1750 0.2611

beta.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.003889 0.020000 0.039468 0.052222 0.727222
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.023333 0.041493 0.060000 0.435556
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.10102 0.05722 1.00000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.6683 0.7439 0.7522 0.8322 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.04961 0.00000 1.00000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.1656 0.2594 0.4983 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4022	0.4356	0.4433	0.4435	0.4511	0.4850

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09389	0.13944	0.13546	0.17167	0.90833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14333	0.13904	0.17556	0.72722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101986	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0055556	0.0083333	0.0778841	0.0245833	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162532 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1522	0.1806	0.1883	0.1879	0.1950	0.2194

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05944	0.10722	0.10924	0.14500	0.87222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06222	0.11111	0.11268	0.15000	0.63000
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060556	0.415464	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0044444	0.2502399	0.4037500	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.130	1.000	0.698	1.000	1.000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07389	0.09167	0.09778	0.09767	0.10333	0.12278

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02500	0.06833	0.07745	0.11111	0.83167
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02667	0.07222	0.08072	0.11667	0.58333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9178	1.0000	0.7931	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2198423	0.1712500	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05111	1.00000	0.67162	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02111	0.04000	0.04500	0.04648	0.05222	0.07778

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02056	0.06111	0.07108	0.10222	0.78667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.06556	0.07474	0.10889	0.57778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.6900 1.0000 0.7679 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2566 0.5696 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.9872 1.0000 0.8412 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.01778 0.03389 0.04222 0.04396 0.05111 0.14333

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01111 0.04389 0.05732 0.08556 0.74889
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.01222 0.04778 0.06103 0.09222 0.54056
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.1750 0.6317 0.5755 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2558 0.5797 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.9261 1.0000 0.8495 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.01000 0.02333 0.02778 0.02913 0.03333 0.07444

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007222	0.038889	0.053446	0.081111	0.765556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.042222	0.057519	0.088889	0.493333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4150	0.4705	0.9928	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8644	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2433	0.4938	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7422	1.0000	0.7955	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01500	0.02778	0.03333	0.03564	0.04111	0.08333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.022778	0.042176	0.062778	0.758333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.025556	0.045881	0.070000	0.513333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.2878	0.4246	0.9372	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9988	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1763	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02833	0.68667	0.56501	0.96611	1.00000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1806	0.2083	0.2189	0.2201	0.2300	0.2972

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.014444	0.033970	0.042222	0.742222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.015556	0.034016	0.046667	0.446667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1558	0.2611	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5506	0.6472	0.6417	0.7406	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.05159 0.00000 1.00000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.001111 0.217310 0.400556 1.000000

```

--- Summary for K = 9 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.3561  0.3883  0.4000  0.4024  0.4139  0.5167

```

```

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.004444 0.017778 0.067142 0.053333 0.789444
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1

```

```

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.004444 0.020000 0.047920 0.056667 0.480556
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.01105 0.00000 1.00000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.2472  0.4111  0.4149  0.5483  1.0000
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.006904 0.000000 1.000000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.02991 0.00000 1.00000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3506	0.3800	0.3883	0.3881	0.3961	0.4239

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09056	0.13500	0.13041	0.16611	0.91222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.13778	0.13286	0.16889	0.77222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102031	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.005000	0.008333	0.077119	0.024583	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162199 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1078	0.1328	0.1394	0.1391	0.1456	0.1672

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05167	0.09778	0.10013	0.13500	0.88556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05444	0.10111	0.10304	0.13889	0.69444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.061667	0.416818	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2501460	0.3637500	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1322	1.0000	0.6994	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02778	0.04056	0.04500	0.04510	0.04944	0.06389

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02833	0.07333	0.07669	0.10944	0.79889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03000	0.07778	0.08019	0.11444	0.62667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9656	1.0000	0.8044	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.234724	0.296528	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1022	1.0000	0.6889	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01500	0.02889	0.03444	0.03593	0.04167	0.10056

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.05056	0.05992	0.08833	0.75722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.05444	0.06352	0.09444	0.65444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.0000 0.4861 0.7878 0.6801 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2751 0.6922 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8701 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.01500 0.03111 0.03667 0.03835 0.04389 0.08056

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01222 0.03944 0.04948 0.07222 0.81000
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01333 0.04333 0.05244 0.07778 0.57556
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.3572 0.4654 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.8617 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2452 0.4778 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9283 1.0000 0.8324 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.003333 0.012222 0.015000 0.015979 0.018889 0.051667

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.009444 0.032778 0.043993 0.063889 0.782222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.010000 0.03556 0.04732 0.07000 0.50222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.4094 0.4784 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    0.86    1.00    1.00    1.00    1.00    1.00
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2268 0.3383 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.7206 1.0000 0.7811 1.0000 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.01222 0.02444 0.02944 0.03073 0.03556 0.07222

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.002778 0.018333 0.033106 0.050556 0.726111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.021111	0.036439	0.056667	0.525556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2596	0.5628	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1061	1.0000	1.0000	0.9986	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1185	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2200	0.8411	0.6429	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.008333	0.017222	0.022222	0.023450	0.028333	0.055000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.023333	0.036039	0.055000	0.707778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.026667	0.040528	0.063333	0.478889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3429	0.7956	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6278	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1653 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.3517 0.8322 0.6583 1.0000 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.05556 0.08611 0.09500 0.09897 0.10944 0.18833

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.002778 0.015000 0.026954 0.039444 0.661111
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003333 0.017778 0.030539 0.045556 0.445556
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.0748 0.0000 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.1133 0.7394 0.8472 0.8320 0.9511 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.03572 0.00000 0.85000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.01833 0.17449 0.33444 1.00000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3572	0.3794	0.3872	0.3876	0.3961	0.4200

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.13833	0.13432	0.16889	0.86500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14111	0.13698	0.17222	0.80889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102236	0.031667	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.005000	0.008333	0.077411	0.024444	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.0005556 0.0061111 0.0150000 0.1625307 0.1188889 1.0000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1689	0.2111	0.2183	0.2185	0.2261	0.2517

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04722	0.09000	0.09689	0.13000	0.93111
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05000	0.09333	0.09915	0.13333	0.67500
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.036111	0.401883	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0027778	0.2367719	0.2611111	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.07889	1.00000	0.68540	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03222	0.04722	0.05222	0.05264	0.05722	0.08278

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02889	0.07417	0.07866	0.11167	0.81667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03000	0.07833	0.08209	0.11667	0.62333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9600	1.0000	0.8024	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.229487	0.209444	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0900	1.0000	0.6855	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01278	0.02111	0.02500	0.02538	0.02944	0.06222

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02278	0.06139	0.06889	0.09722	0.81944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.06556	0.07283	0.10444	0.57889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.7506 1.0000 0.7752 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2713 0.6944 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000  1.000  1.000  0.862  1.000  1.000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.002778 0.010000 0.013333 0.014263 0.017778 0.068333

beta.ar:
Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.02000 0.05333 0.06204 0.08778 0.79833
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00000 0.02222 0.05778 0.06660 0.09500 0.58556
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.7333 1.0000 0.7873 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2911 0.8629 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 1.0000 1.0000 0.8973 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.003333 0.013333 0.016667 0.017638 0.020556 0.074444

```


beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01056	0.04000	0.05169	0.07556	0.78167

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.04444	0.05573	0.08333	0.55111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.07444	0.68500	0.57611	1.00000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8578	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2694	0.6467	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8933	1.0000	0.8430	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001667	0.007778	0.010000	0.010877	0.012778	0.055000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006111	0.028333	0.040526	0.060000	0.749444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.032222	0.044602	0.067778	0.531111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.7944	0.5892	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4206	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2063	0.0325	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4283	0.9556	0.7035	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01556	0.02722	0.03389	0.03544	0.04111	0.18000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002778	0.021111	0.034941	0.053333	0.733333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.024444	0.039098	0.061111	0.501667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4917	0.4431	0.8128	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1217	0.9333	0.9828	0.9570	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.09484 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.3461  0.3914  0.7456  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.01889 0.03444 0.03944 0.04119 0.04667 0.12111

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.001667 0.013333 0.028533 0.042222 0.688333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.002222 0.015556 0.032147 0.048889 0.475556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.0000  0.2705  0.5844  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.8400  0.9306  0.8972  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.06796 0.000000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.01222 0.29155 0.59833 1.00000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3528	0.3794	0.3883	0.3877	0.3956	0.4289

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09333	0.13778	0.13466	0.17000	0.90611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14111	0.13746	0.17333	0.75111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102025	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0769476	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162455 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1083	0.1300	0.1361	0.1359	0.1417	0.1628

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04944	0.09833	0.10178	0.13833	0.83278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05222	0.10222	0.10477	0.14333	0.69000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.062778	0.417127	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2500556	0.3872222	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1350	1.0000	0.6996	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03222	0.04611	0.05056	0.05138	0.05556	0.10056

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02889	0.07556	0.08104	0.11556	0.88944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03000	0.08000	0.08436	0.12111	0.66778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7478	0.8917	0.7531	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.002222	0.245932	0.349583	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2528	1.0000	0.7206	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01000	0.02111	0.02500	0.02609	0.03000	0.07167

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01500	0.05278	0.06338	0.09222	0.81444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.05667	0.06681	0.09778	0.72556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000 0.7244 1.0000 0.7713 1.0000 1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2705	0.7035	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8627	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.008333	0.020556	0.025556	0.027432	0.031667	0.107222

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01278	0.04389	0.05485	0.07944	0.82167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.04778	0.05834	0.08556	0.61056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3961	0.9211	0.6889	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9033	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2475	0.5001	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7411	1.0000	0.7967	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.008333	0.015556	0.019444	0.020940	0.025000	0.049444

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.007778 0.034444 0.048911 0.072222 0.813889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.008889 0.037778 0.052532 0.078889 0.541667
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.2633 0.4409 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.7067 1.0000 1.0000 0.9999 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2388 0.4354 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.8444 1.0000 0.8143 1.0000 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.008333 0.019444 0.024444 0.025892 0.031111 0.078889

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.003889 0.025556 0.042564 0.065000 0.766111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :

```


Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.028889	0.046565	0.073333	0.563333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3250	0.4338	0.8817	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8161	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2070	0.3076	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6178	0.9050	0.7410	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.006111	0.015556	0.019444	0.020647	0.023889	0.065556

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.026667	0.041085	0.060000	0.725556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.030000	0.045625	0.068889	0.454444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2927	0.6861	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5361	1.0000	1.0000	0.9995	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1456 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.1633 0.7439 0.5967 0.9544 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.01056 0.01889 0.02333 0.02449 0.02833 0.05667

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.002778 0.021111 0.036561 0.053333 0.686111
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003333 0.024444 0.040996 0.062222 0.452222
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.3142 0.7261 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9972 1.0000 0.9839 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1148 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.6011 0.5094 0.8633 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3533	0.3794	0.3878	0.3875	0.3956	0.4300

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.13889	0.13634	0.17167	0.90833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14222	0.13929	0.17500	0.73000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101938	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0773253	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162327 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1189	0.1378	0.1450	0.1448	0.1517	0.1728

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05556	0.10167	0.10564	0.14111	0.83667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05778	0.10556	0.10860	0.14556	0.68167
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060000	0.415315	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2484772	0.3916667	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1256	1.0000	0.6978	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02056	0.04000	0.04444	0.04476	0.04944	0.06444

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03889	0.08111	0.08768	0.11889	0.83278
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04111	0.08556	0.09126	0.12444	0.63833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9622	1.0000	0.8044	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.233252	0.251111	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1011	1.0000	0.6896	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.006667	0.017778	0.021111	0.021358	0.025000	0.054444

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02389	0.06500	0.07299	0.10278	0.84889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.07000	0.07689	0.10944	0.59500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.7972 1.0000 0.7893 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2704 0.7153 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8629 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.004444 0.014444 0.018333 0.020372 0.023889 0.095556

beta.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.01556 0.05028 0.06099 0.08722 0.79111
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.01667 0.05444 0.06524 0.09444 0.55389
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.4844 0.8589 0.6958 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2788 0.7306 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8856 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.007778 0.018333 0.022778 0.023996 0.028889 0.072222

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.008889 0.032778 0.047171 0.068333 0.792778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.010000 0.03667 0.05052 0.07444 0.56944
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.3561 0.4630 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.4028 1.0000 1.0000 0.9999 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2027 0.1399 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.3811 0.8889 0.6800 1.0000 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.007222 0.015000 0.018333 0.020138 0.023889 0.065556

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.005000 0.030000 0.04535 0.06722 0.79333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.033333	0.049547	0.075417	0.553333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.2061	0.4131	0.9267	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.715	1.000	1.000	1.000	1.000	1.000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2002	0.1799	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6661	0.9878	0.7644	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01722	0.03500	0.04111	0.04273	0.04944	0.09500

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.008889	0.027155	0.037778	0.712222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.010000	0.029735	0.043333	0.530556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1479	0.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8706	0.9611	0.9167	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.00000 0.00000 0.00000 0.07229 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.6189  0.5245  1.0000  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.01056 0.02500 0.03111 0.03296 0.03944 0.08333

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005556 0.024444 0.039610 0.056667 0.683889
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.006667 0.028889 0.044668 0.065556 0.547778
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.0000  0.1447  0.1850  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  1.0000  1.0000  0.9899  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.06679 0.000000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.3006  0.3491  0.6272  1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3522	0.3794	0.3878	0.3875	0.3956	0.4272

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09722	0.14000	0.13710	0.17222	0.90778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1006	0.1433	0.1402	0.1756	0.7611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102072	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774610	0.0250000	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162452 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1100	0.1322	0.1378	0.1382	0.1444	0.1683

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05611	0.10278	0.10720	0.14278	0.88278
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05792	0.10667	0.11019	0.14778	0.62556
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.062222	0.416602	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2508117	0.4062500	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1344	1.0000	0.6994	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02889	0.05000	0.05556	0.05598	0.06111	0.10889

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03833	0.08167	0.08770	0.12056	0.83889
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04111	0.08667	0.09148	0.12667	0.67833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5272	0.6572	0.6627	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.002778	0.257789	0.461667	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4572	1.0000	0.7622	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.007778	0.017778	0.020556	0.021346	0.025000	0.043333

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02889	0.06778	0.07593	0.10542	0.77667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.07222	0.08007	0.11222	0.55722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.8778 1.0000 0.8005 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2728 0.7496 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8526 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.008333 0.017778 0.022222 0.023401 0.027222 0.095000

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.04500 0.05727 0.08389 0.80611
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01333 0.04889 0.06104 0.09056 0.62500
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.3272 0.7956 0.6477 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9472 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2681 0.6390 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8752 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00500 0.01111 0.01500 0.01574 0.01944 0.07722

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01056	0.04111	0.05397	0.07833	0.78222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01111	0.04556	0.05810	0.08556	0.61222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1183	0.7322	0.5927	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2786	0.7536	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.985	1.000	0.862	1.000	1.000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.006667	0.014444	0.018333	0.019191	0.022778	0.045000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.026111	0.042224	0.062222	0.787222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.029167	0.045796	0.070000	0.634444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.006111	0.375046	0.945556	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	1.000	1.000	0.999	1.000	1.000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	0.000	0.162	0.000	1.000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3578	0.9467	0.6883	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01056	0.02278	0.02944	0.03086	0.03722	0.07333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002778	0.022500	0.039991	0.060000	0.700556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.025556	0.044559	0.068889	0.472778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2884	0.6639	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9986	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1384 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.1894 0.6878 0.5691 0.8956 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.007778 0.017778 0.023333 0.023871 0.028889 0.052222

beta.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003889 0.023056 0.039223 0.057222 0.675556
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1 1 1 1 1 1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1 1 1 1 1 1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1 1 1 1 1 1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
1 1 1 1 1 1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.026667 0.044189 0.066667 0.473333
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2003 0.3633 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.9881 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.09694 0.00000 1.00000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.1711 0.6794 0.5767 0.9728 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3544	0.3789	0.3856	0.3866	0.3944	0.4239

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09722	0.14000	0.13766	0.17222	0.90056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1000	0.1439	0.1410	0.1761	0.7222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102035	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774961	0.0240278	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162480 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1017	0.1294	0.1356	0.1355	0.1417	0.1600

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06167	0.10833	0.11173	0.14667	0.88556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06444	0.11222	0.11506	0.15111	0.65722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.062222	0.417156	1.000000	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.250889	0.407083	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1322	1.0000	0.6999	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04778	0.06500	0.07056	0.07125	0.07722	0.10000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02500	0.06833	0.07718	0.11000	0.85056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02667	0.07222	0.08029	0.11556	0.58444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9356	1.0000	0.7950	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2229156	0.1795833	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05722	1.00000	0.67486	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.007778	0.020000	0.023889	0.024284	0.027222	0.082778

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02889	0.06944	0.07764	0.10778	0.82667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03111	0.07444	0.08200	0.11556	0.65278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000 0.7022 0.9572 0.7590 1.0000 1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2817	0.7793	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8729	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.006667	0.015000	0.019444	0.020921	0.024444	0.078889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.020000	0.05722	0.06765	0.09556	0.78667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.02222	0.06222	0.07218	0.10333	0.56722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4633	0.8556	0.6906	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2718	0.7231	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8855	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.003889	0.010000	0.012778	0.013680	0.016111	0.072778

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.04500	0.05758	0.08444	0.69778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04917	0.06247	0.09222	0.50778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2228	0.8411	0.6349	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2802	0.7917	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8761	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.005556	0.012778	0.016111	0.017024	0.020000	0.075000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.036667	0.052233	0.075556	0.766111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.041111	0.056328	0.084444	0.558333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.5300	0.5135	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7617	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2080	0.2378	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5928	0.9450	0.7409	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.009444	0.019444	0.023889	0.025360	0.029444	0.061667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.028333	0.045470	0.067222	0.709444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.032778	0.050100	0.076667	0.509444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1339	0.3594	0.7911	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4139	1.0000	1.0000	0.9978	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1622 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.1822 0.6894 0.5828 0.9656 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.1928 0.2156 0.2233 0.2250 0.2328 0.3000

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.015000 0.036992 0.040556 0.706667
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.004444 0.016667 0.034642 0.044444 0.470556
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1164 0.0000 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.3006 0.4383 0.4183 0.5617 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.02355 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.1299 0.1022 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3467	0.3800	0.3872	0.3875	0.3961	0.4272

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09611	0.14056	0.13703	0.17222	0.91444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1000	0.1444	0.1406	0.1767	0.7028

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102109	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0777331	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162335 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1094	0.1356	0.1417	0.1419	0.1483	0.1683

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05944	0.10667	0.10998	0.14611	0.88889
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06222	0.11056	0.11338	0.15111	0.63222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060556	0.416070	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.004444	0.250928	0.378194	1.000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1306	1.0000	0.6988	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04222	0.05722	0.06278	0.06371	0.06833	0.12389

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03000	0.07667	0.08372	0.11889	0.82611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03222	0.08111	0.08741	0.12444	0.65556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5578	0.7050	0.6776	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001667	0.250879	0.401667	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4111	1.0000	0.7494	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01111	0.02222	0.02611	0.02665	0.03056	0.05056

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01722	0.05722	0.06799	0.09889	0.79722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01889	0.06111	0.07182	0.10556	0.57556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.7933 1.0000 0.7854 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2529 0.5674 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 1.0000 1.0000 0.8411 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.008889 0.018333 0.022222 0.023944 0.027778 0.101667

beta.ar:
Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.01333 0.04944 0.06156 0.09000 0.79778
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.000000 0.01444 0.05333 0.06541 0.09778 0.52556
Modality 2 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.2444 0.7067 0.6087 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 0.0000 0.0000 0.2567 0.5663 1.0000
Modality 5 :
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.0000 1.0000 1.0000 0.8736 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median Mean 3rd Qu. Max.
0.01056 0.02333 0.02833 0.03029 0.03444 0.09167

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.002778 0.025556 0.044724 0.067778 0.744444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.027778 0.047799 0.074444 0.574444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.6544 0.5537 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.3639 1.0000 1.0000 0.9991 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2091 0.1842 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.3161 0.8439 0.6585 1.0000 1.0000

```

--- Summary for K = 7 ---

```

Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.006667 0.019444 0.025000 0.026136 0.031111 0.070556

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006111 0.032222 0.048657 0.071667 0.748889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.035556	0.052751	0.080000	0.567778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.5339	0.4897	0.9050	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6894	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1920	0.1636	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4661	0.8389	0.6815	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.007222	0.017222	0.021667	0.022597	0.027222	0.050000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.021667	0.038519	0.056667	0.707222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.025556	0.043036	0.065556	0.488889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2066	0.3167	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9954	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1011 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.7011 0.5614 1.0000 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.02167 0.03500 0.04222 0.04530 0.05222 0.12778

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.001667 0.015000 0.034255 0.050556 0.668889
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.002222 0.017778 0.038596 0.058889 0.463333
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2413 0.4933 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.8906 0.9611 0.9293 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.05967 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2581 0.5256 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2167	0.2456	0.2517	0.2523	0.2594	0.2822

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09167	0.13500	0.13099	0.16667	0.92000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.13778	0.13351	0.17000	0.76000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102028	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0778827	0.0245833	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162458 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04278	0.05944	0.06444	0.06489	0.07000	0.08556

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05167	0.09778	0.09968	0.13500	0.85556
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05444	0.10139	0.10263	0.13944	0.69667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.058333	0.414613	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2480049	0.3718056	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1250	1.0000	0.6976	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01389	0.02889	0.03333	0.03472	0.03944	0.07889

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.06500	0.07173	0.10333	0.90222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.06889	0.07477	0.10778	0.60556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5672	0.7567	0.6895	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.241369	0.367778	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3589	1.0000	0.7317	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00500	0.01222	0.01444	0.01528	0.01778	0.05000

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.05000	0.05906	0.08611	0.86556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.05333	0.06227	0.09111	0.59333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.6839 1.0000 0.7637 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2677 0.7203 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8602 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.006111 0.015000 0.020000 0.020353 0.023889 0.058889

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.01278 0.04222 0.05212 0.07500 0.78722
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.01444 0.04556 0.05509 0.08000 0.52667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000 0.000 0.310 0.445 1.000 1.000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.8089 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2165 0.2525 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9633 1.0000 0.8380 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0050000 0.0066667 0.0074596 0.0088889 0.0555556

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01167	0.03833	0.04877	0.07111	0.78611
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.04222	0.05273	0.07778	0.55444
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09111	0.79333	0.60371	1.00000	1.00000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2837	0.8439	1.0000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8696	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.007778	0.010556	0.010738	0.012778	0.023889

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.015556	0.030466	0.047222	0.765556
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002222	0.017778	0.033052	0.052222	0.476667

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4961	0.4839	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02444	1.00000	1.00000	0.99691	1.00000	1.00000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1452	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.7122	0.5633	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.006667	0.013333	0.016667	0.017559	0.021111	0.036667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.005556	0.020687	0.029444	0.738889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.006667	0.022996	0.033333	0.427222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2012	0.2067	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9472	1.0000	0.9362	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.05282 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.2928  0.4353  1.0000  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.01778 0.03167 0.03667 0.03790 0.04222 0.09500

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.001667 0.007222 0.018053 0.023333 0.705000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.002222 0.007778 0.019822 0.025972 0.424444
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.0000  0.1261  0.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.3689  0.5567  0.5269  0.7122  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.02327 0.000000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.0000  0.0000  0.2732  0.5683  1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2244	0.2450	0.2517	0.2523	0.2600	0.2844

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09389	0.13722	0.13360	0.16833	0.94889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09667	0.14000	0.13628	0.17111	0.76889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102097	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.005000	0.008333	0.077258	0.024583	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162402 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04722	0.06111	0.06556	0.06591	0.07056	0.09111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05556	0.10111	0.10404	0.13889	0.91278

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05778	0.10444	0.10687	0.14333	0.68611

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.057778	0.414293	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0044444	0.2486465	0.3727778	1.0000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1222	1.0000	0.6972	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01222	0.02278	0.02667	0.02707	0.03111	0.05389

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.06667	0.07304	0.10556	0.84056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.07000	0.07587	0.11056	0.75222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8361	1.0000	0.7875	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2284970	0.2244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08944	1.00000	0.68236	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.002222	0.011111	0.013889	0.014583	0.017222	0.043333

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.05667	0.06534	0.09222	0.85667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.06000	0.06888	0.09778	0.57444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.0000 0.7561 1.0000 0.7807 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2612 0.6126 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8306 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.004444 0.010556 0.013889 0.014882 0.017778 0.054444

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.006667 0.035556 0.048611 0.073889 0.790000
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.006667 0.037778 0.051485 0.078889 0.602778
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.1867 0.7239 0.6050 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.8683 1.0000 1.0000 1.0000 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2299 0.3446 1.0000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9222 1.0000 0.8393 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.001111 0.005000 0.006667 0.007374 0.008889 0.036111

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.04056	0.05186	0.07278	0.74889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04444	0.05571	0.08000	0.53944

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.5517	0.5127	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2524	0.5569	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9661	1.0000	0.8425	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.006111	0.007778	0.008233	0.010000	0.018333

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.022778	0.037614	0.056111	0.748333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.025556	0.041087	0.062778	0.537222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2942	0.7056	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1211	1.0000	1.0000	0.9989	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1471	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5022	1.0000	0.7307	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.003333	0.008333	0.010556	0.011260	0.013333	0.026667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.022778	0.035876	0.051667	0.731111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.025556	0.039723	0.058889	0.485556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1970	0.2672	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9962	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.00000 0.00000 0.00000 0.09569 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01389 0.82222 0.61557 1.00000 1.00000

--- Summary for K = 9 ---
Z.ar:
    Min.    1st Qu.    Median      Mean    3rd Qu.      Max.
0.0005556 0.0050000 0.0072222 0.0073040 0.0094444 0.0211111

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00500 0.02278 0.03770 0.05278 0.72611
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005556 0.026667 0.042070 0.062222 0.500000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2607 0.5928 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.320 1.000 1.000 0.999 1.000 1.000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1125 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.9017 0.6314 1.0000 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2194	0.2450	0.2522	0.2527	0.2594	0.2850

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0950	0.1389	0.1352	0.1700	0.9239

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09778	0.14222	0.13806	0.17333	0.73778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102147	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.078190	0.024028	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162432 0.117778 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04000	0.05667	0.06111	0.06099	0.06556	0.07944

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05556	0.10167	0.10544	0.14056	0.89000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05778	0.10556	0.10824	0.14444	0.69111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060000	0.415793	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0055556	0.2493771	0.3769444	1.0000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1289	1.0000	0.6985	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.009444	0.021111	0.024444	0.024824	0.028333	0.044444

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.07222	0.07851	0.11111	0.87278

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02889	0.07611	0.08171	0.11667	0.60889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9378	1.0000	0.7980	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2262954	0.1736111	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06722	1.00000	0.67911	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.002222	0.011111	0.013889	0.014650	0.017222	0.058889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02347	0.06000	0.06964	0.09778	0.83111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02556	0.06444	0.07317	0.10389	0.66056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5644 0.8789 0.7161 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2748 0.6607 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8701 1.0000 1.0000

```

--- Summary for K = 5 ---

Z.ar:

```

      Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.003333 0.007778 0.010000 0.010938 0.012778 0.061111

```

beta.ar:

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01444 0.04500 0.05667 0.08111 0.81722
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

```

gamma.ar:

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01556 0.04889 0.06039 0.08778 0.70000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.3339 0.8789 0.6657 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2825 0.8436 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8846 1.0000 1.0000

```

--- Summary for K = 6 ---

Z.ar:

```

      Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0044444 0.0061111 0.0066622 0.0077778 0.0461111

```



```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01222 0.03722 0.04926 0.06889 0.73722
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01444 0.04111 0.05321 0.07556 0.55000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.6900  0.5395  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.9467  1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2434  0.4363  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.8344  1.0000  0.8041  1.0000  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.001667 0.005556 0.007778 0.007989 0.010000 0.017778

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005556 0.025000 0.039213 0.056111 0.751111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.028889	0.042689	0.062778	0.480000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3453	0.9767	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9984	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1355	0.0000	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003889	0.930000	0.641091	1.000000	1.000000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.004444	0.006111	0.006172	0.007778	0.016667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.026667	0.041085	0.057778	0.753333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.031111	0.045411	0.065556	0.494444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3621	0.9794	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9992	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1396 0.0000 1.0000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.05833 0.94000 0.65034 1.00000 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.002222 0.007222 0.010000 0.010361 0.012778 0.026667

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.026667 0.043231 0.058333 0.715000
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
  1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
  1         1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.007778 0.031111 0.047342 0.067778 0.507778
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2213 0.4133 1.0000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.1167 1.0000 1.0000 0.9998 1.0000 1.0000
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1007 0.0000 1.0000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.4328 0.4537 0.9161 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2100	0.2456	0.2522	0.2522	0.2594	0.2883

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09444	0.13833	0.13481	0.17000	0.92278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09778	0.14111	0.13768	0.17333	0.75722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102054	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0774118	0.0238889	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162489 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04000	0.05833	0.06222	0.06264	0.06722	0.08389

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05667	0.10278	0.10752	0.14278	0.88222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05889	0.10667	0.11045	0.14722	0.64056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.059444	0.415498	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2484605	0.3427778	1.0000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1261	1.0000	0.6983	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01333	0.02444	0.02778	0.02840	0.03167	0.07389

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03278	0.07333	0.08195	0.11222	0.84056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03444	0.07694	0.08526	0.11778	0.61889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7150	0.9144	0.7494	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001111	0.235039	0.263889	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2150	1.0000	0.7065	1.0000	1.0000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.003333	0.009444	0.011667	0.011941	0.013889	0.043333

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01944	0.05667	0.06684	0.09556	0.81611

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.06056	0.07027	0.10111	0.61000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.6606 1.0000 0.7543 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2667 0.6944 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8656 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.002222 0.007778 0.010000 0.010578 0.012222 0.063333

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.01889 0.05056 0.06097 0.08500 0.81722
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.02111 0.05444 0.06463 0.09111 0.58222
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2039 0.7883 0.6241 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2584 0.5769 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8733 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.001111 0.004444 0.006111 0.006666 0.008333 0.030000

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01403	0.04278	0.05599	0.07833	0.73611

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01556	0.04722	0.06000	0.08556	0.65833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4367	0.4868	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6678	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2576	0.6699	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9900	1.0000	0.8386	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0066667	0.0094444	0.0099511	0.0122222	0.0572222

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.03778	0.05072	0.07333	0.77111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01111	0.04222	0.05522	0.08208	0.53111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4917	0.5126	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9156	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2206	0.3306	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7867	1.0000	0.8038	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001667	0.005556	0.007778	0.008362	0.010000	0.031111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005694	0.029444	0.046287	0.065000	0.744444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.033333	0.050451	0.073333	0.468889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3750	0.9556	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6850	1.0000	1.0000	0.9999	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1809 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.3372 0.9556 0.6875 1.0000 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.003333 0.008333 0.010556 0.011130 0.013333 0.025556

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.002222 0.017222 0.033273 0.048889 0.673889
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.002222 0.020000 0.037553 0.056667 0.451111
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2860 0.6817 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.002778 1.000000 1.000000 0.993027 1.000000 1.000000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.1018 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.1106 0.3459 0.7344 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

Automated Report

Hyperparameters

Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2211	0.2456	0.2528	0.2526	0.2594	0.2811

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09556	0.14000	0.13719	0.17111	0.89444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09889	0.14333	0.14022	0.17556	0.73722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102109	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.005000	0.008333	0.077355	0.025000	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162391 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04111	0.05722	0.06167	0.06168	0.06611	0.08167

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05944	0.10500	0.10932	0.14333	0.86056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06222	0.10889	0.11233	0.14778	0.65222

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060000	0.415960	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2502579	0.3606944	1.0000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1272	1.0000	0.6984	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.009444	0.019444	0.022222	0.022604	0.025556	0.037222

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03167	0.07222	0.08103	0.11167	0.85778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03333	0.07611	0.08409	0.11667	0.60389

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8378	1.0000	0.7877	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2261193	0.2140278	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08833	1.00000	0.68217	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.007222	0.013889	0.016667	0.017625	0.021111	0.041111

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01556	0.05222	0.06334	0.09167	0.77556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.05556	0.06652	0.09667	0.59111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.8678 1.0000 0.7921 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2578 0.5613 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9178 1.0000 0.8119 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.002778 0.009444 0.012222 0.013340 0.015556 0.053889

beta.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.01500 0.04611 0.05803 0.08167 0.79500
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.01667 0.05000 0.06142 0.08778 0.67667
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.3722 1.0000 0.6984 1.0000 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2582 0.5967 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.9333 1.0000 0.8261 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.001667 0.005556 0.007222 0.007940 0.009444 0.051111

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01278	0.04333	0.05603	0.07944	0.74444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.04778	0.06026	0.08667	0.54333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.70278	0.56996	1.00000	1.00000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2717	0.6892	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9075	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.002222	0.006667	0.008333	0.009133	0.011111	0.027222

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.009444	0.033889	0.047522	0.067778	0.740000

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.03778	0.05193	0.07556	0.58500

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3856	0.9456	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8956	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.187303	0.001806	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6683	1.0000	0.7680	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0027778	0.0038889	0.0041040	0.0050000	0.0100000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.026667	0.041378	0.059444	0.746111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.006667	0.030556	0.045506	0.067778	0.518889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.7372	0.5551	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.9995	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.


```

0.0000 0.0000 0.0000 0.1638 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.2417 0.9544 0.6748 1.0000 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.001111 0.004444 0.005556 0.005988 0.007222 0.012778

beta.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.022222 0.037133 0.053333 0.671111
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.005556 0.026667 0.041831 0.062222 0.490556
  Modality 2 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2435 0.5072 1.0000
  Modality 3 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000 1.000 1.000 0.994 1.000 1.000
  Modality 4 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.09724 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.7406 0.5641 1.0000 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

iClusterBayes Summary Report

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Key Summary Statistics

--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2194	0.2456	0.2528	0.2524	0.2589	0.2850

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09722	0.14111	0.13775	0.17222	0.89333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1000	0.1450	0.1412	0.1761	0.7489

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.102049	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0083333	0.0776963	0.0244444	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162454 0.118889 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04056	0.05556	0.06056	0.06028	0.06444	0.08222

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0650	0.1094	0.1137	0.1467	0.8544

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06778	0.11333	0.11701	0.15222	0.69333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.061111	0.416666	1.000000	1.000000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001111	0.005000	0.250839	0.386806	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1306	1.0000	0.6992	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.008333	0.019444	0.022222	0.022305	0.025556	0.038889

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03167	0.07222	0.08050	0.11222	0.78889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03444	0.07667	0.08383	0.11778	0.55444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8900	1.0000	0.7919	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2208022	0.1716667	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05778	1.00000	0.67531	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.003333	0.010000	0.012778	0.013227	0.016111	0.052222

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02222	0.06111	0.07171	0.10111	0.85722

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02444	0.06556	0.07519	0.10778	0.64944

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.5994 1.0000 0.7435 1.0000 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2666 0.6369 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8654 1.0000 1.0000

--- Summary for K = 5 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.003333 0.008889 0.012222 0.013323 0.016111 0.056111

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.021111 0.053333 0.06548 0.09000 0.82167
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.023333 0.05778 0.06911 0.09667 0.67222
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.02778 0.64000 0.55887 1.00000 1.00000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2574 0.6028 1.0000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.8728 1.0000 1.0000

--- Summary for K = 6 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.001111 0.004444 0.006667 0.007253 0.008889 0.051111

```

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.04833	0.06047	0.08500	0.75611

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.05333	0.06527	0.09333	0.51778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2828	0.9450	0.6609	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2799	0.8483	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8725	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0038889	0.0055556	0.0061049	0.0072222	0.0311111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.03667	0.05127	0.07222	0.74722

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
--------------	--	--	--	--	--

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.04111	0.05595	0.08111	0.53278

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.4106	0.4806	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6494	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2196	0.2267	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7761	1.0000	0.7986	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.003333	0.009444	0.012778	0.013408	0.016667	0.031667

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002222	0.021111	0.040231	0.060556	0.734444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.024444	0.044021	0.068889	0.514444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2938	0.6828	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4700	1.0000	1.0000	0.9993	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1439 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.05222 0.68500 0.56825 0.97389 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.003333 0.010000 0.012222 0.013128 0.015556 0.036111

beta.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.002778 0.021111 0.038919 0.056111 0.659444
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
  1      1      1      1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.000000 0.003333 0.025556 0.043236 0.065000 0.487778
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 0.0000 0.0000 0.2349 0.4800 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000 1.0000 1.0000 0.9836 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.07875 0.00000 1.00000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.07222 0.31343 0.63167 1.00000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.

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--- Summary for K = 1 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2194	0.2439	0.2517	0.2517	0.2589	0.2889

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09833	0.14111	0.13813	0.17222	0.88278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1022	0.1450	0.1418	0.1767	0.7239

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.101992	0.031111	1.000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005000	0.008333	0.077249	0.023889	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.006111 0.015000 0.162307 0.118333 1.000000

--- Summary for K = 2 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04278	0.05944	0.06333	0.06315	0.06722	0.08722

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06333	0.10833	0.11277	0.14722	0.89667
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06625	0.11278	0.11623	0.15222	0.67167
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.001667	0.060000	0.415376	1.000000	1.000000
Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0005556	0.0050000	0.2483599	0.3805556	1.0000000
Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1294	1.0000	0.6982	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01444	0.02611	0.03056	0.03089	0.03500	0.05833

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02667	0.07056	0.07914	0.11278	0.85222
Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1
Modality 3 :					

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.07444	0.08240	0.11778	0.60833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9189	1.0000	0.7933	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0005556	0.2238627	0.1551389	1.0000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05389	1.00000	0.67336	1.00000	1.00000

--- Summary for K = 4 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.002222	0.007778	0.009444	0.010160	0.012222	0.030556

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03056	0.06667	0.07649	0.10389	0.82667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03333	0.07111	0.08063	0.11111	0.68667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

0.0000 0.9361 1.0000 0.7971 1.0000 1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2543	0.5308	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8425	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0050000	0.0072222	0.0076276	0.0094444	0.0466667

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.05778	0.06776	0.09611	0.76111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.06333	0.07259	0.10389	0.59111

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6694	1.0000	0.7711	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2873	0.9133	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8982	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.006667	0.008333	0.008843	0.010556	0.027778

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01278	0.04667	0.05972	0.08556	0.79111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01444	0.05111	0.06395	0.09333	0.57056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3372	0.4553	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8917	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2276	0.3649	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9194	1.0000	0.8313	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0005556	0.0044444	0.0061111	0.0067271	0.0083333	0.0250000

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01056	0.03778	0.05240	0.07444	0.75944

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.04222	0.05722	0.08333	0.52333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0500	0.4181	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.204194	0.008194	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6533	1.0000	0.7649	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001111	0.005556	0.007778	0.008400	0.010556	0.034444

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.036667	0.053713	0.076111	0.763333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.04222	0.05841	0.08667	0.50833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1400	0.4153	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4972	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

```

0.0000 0.0000 0.0000 0.1802 0.0000 1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.3544 0.9178 0.6819 1.0000 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.003333 0.008333 0.010556 0.010679 0.012778 0.027222

beta.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.001667 0.016111 0.034721 0.050000 0.671667
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.000000 0.002222 0.018889 0.038832 0.058333 0.446667
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.2846 0.6339 1.0000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 1.0000 1.0000 0.9865 1.0000 1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.06503 0.00000 1.00000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.3939 0.4465 0.9556 1.0000

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1-9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.